

GLAD-M35: a joint *P* and *S* global tomographic model with uncertainty quantification

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SUMMARY

We present our third and final generation joint *P* and *S* global adjoint tomography (GLAD) model, GLAD-M35, and quantify its uncertainty based on a low-rank approximation of the inverse Hessian. Starting from our second-generation model, GLAD-M25, we added 680 new earthquakes to the database for a total of 2160 events. New *P*-wave categories are included to compensate for the imbalance between *P*- and *S*-wave measurements, and we enhanced the window selection algorithm to include more major-arc phases, providing better constraints on the structure of the deep mantle and more than doubling the number of measurement windows to 40 million. Two stages of a Broyden–Fletcher–Goldfarb–Shanno (BFGS) quasi-Newton inversion were performed, each comprising five iterations. With this BFGS update history, we determine the model's standard deviation and resolution length through randomized singular value decomposition.

Key words: Computational seismology; Seismic tomography; Waveform inversion; Wave propagation.

1 INTRODUCTION

For nearly half a century, global seismic tomographic models have been constructed using various aspects of seismic data (e.g. Sen Gupta & Toksöz 1976; Aki *et al.* 1977; Dziewonski *et al.* 1977), including body-wave traveltimes (e.g. Dziewonski 1984; Zhou *et al.* 2006), surface-wave dispersion (e.g. Trampert & Woodhouse 1995; Ekström 2011), long-period free oscillations (e.g. He & Tromp 1996) and *S* and surface waveforms (e.g. Li & Romanowicz 1996). While these studies generally agree on the long-wavelength features of the mantle, further increasing the resolution has proven to be challenging, and there are disagreements and even contradictory results for fine-scale mantle structure in different models. Several factors contribute to this problem. First-order corrections are often used to handle depth variation of the Moho, assuming good knowledge of the crust, which can contaminate mantle structure (e.g. Bozdağ & Trampert 2008; Ferreira *et al.* 2010). Secondly, utilizing only certain aspects of the data limits sensitivity to mantle heterogeneity.

To address these limitations, adjoint-based tomography numerically solves the wave equation and obtains a model by iteratively

minimizing differences between observed and synthetic seismograms (e.g. Tarantola 1984; Virieux & Operto 2009; Tromp 2020). The main challenge for global adjoint tomography is the computational cost. For each event, a 3-D simulation of seismic wave propagation is needed. To account for attenuation, we have the choice of using certain approximations (Karaoğlu & Romanowicz 2018) or saving snapshots of the 3-D wavefield (Komatitsch *et al.* 2016). In most cases, the computational cost of wave simulations is the main bottleneck, but on exascale high-performance computing (HPC) clusters, the cost of reading and writing large wavefield snapshots becomes the main burden.

P and *S* data sets available for tomography can be very different, and most of the earlier tomographic models focused on either *P* or *S* wave speed. Su & Dziewonski (1997) presented one of the first joint *P* and *S* models by compiling waveform and traveltime data. The inferred anticorrelation between bulk sound speed and *S* wave speed in the deep mantle sheds light on the thermochemistry of the region. Later studies using free oscillation (Ishii & Tromp 1999) and traveltimes (Masters *et al.* 2000) confirmed this discovery. Other studies (e.g. Kennett *et al.* 1998; Simmons *et al.* 2010) presented

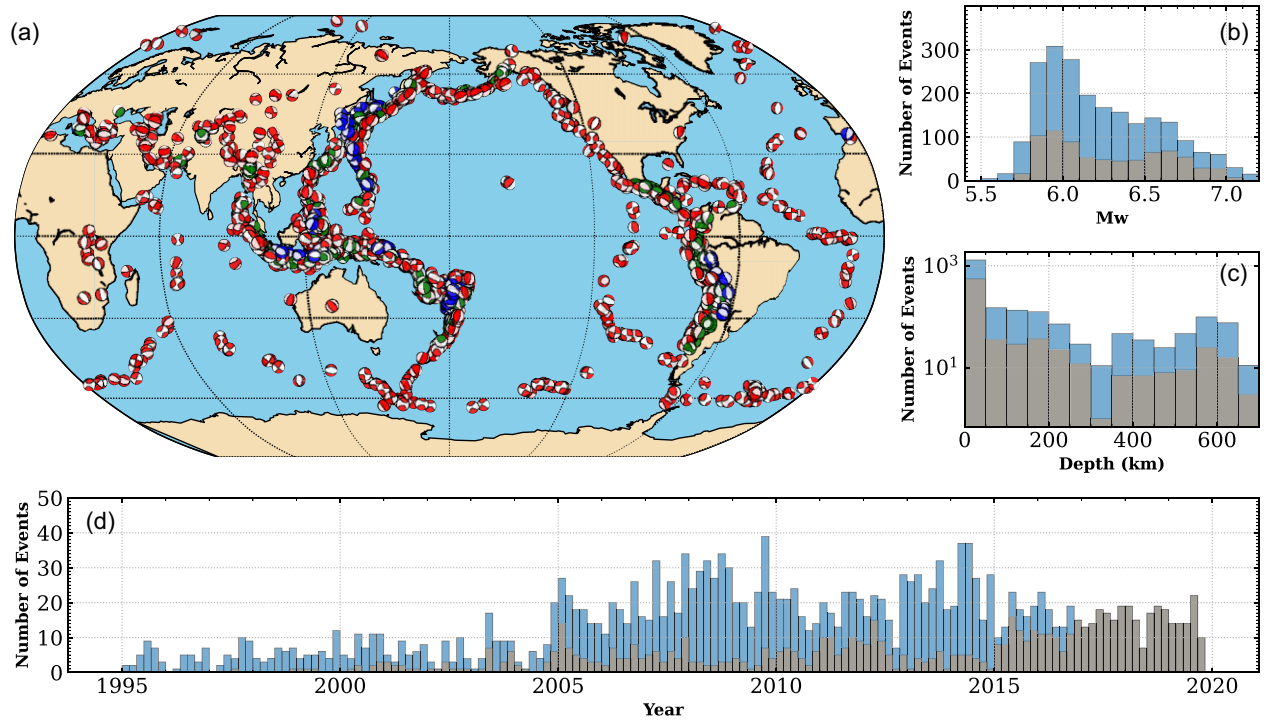


Figure 1. Distribution of the 2160 earthquakes used to construct model GLAD-M35. Grey shading in the histograms indicates newly added data from 680 earthquakes. (a) The colour of each beach ball corresponds to earthquake depth. Red: <50 km. Green: $50\text{--}300$ km. Blue: >300 km. (b) The number of events binned by magnitude M_w . (c) The number of events binned by hypocentre depth (plotted on a log scale). (d) The number of events binned by year.

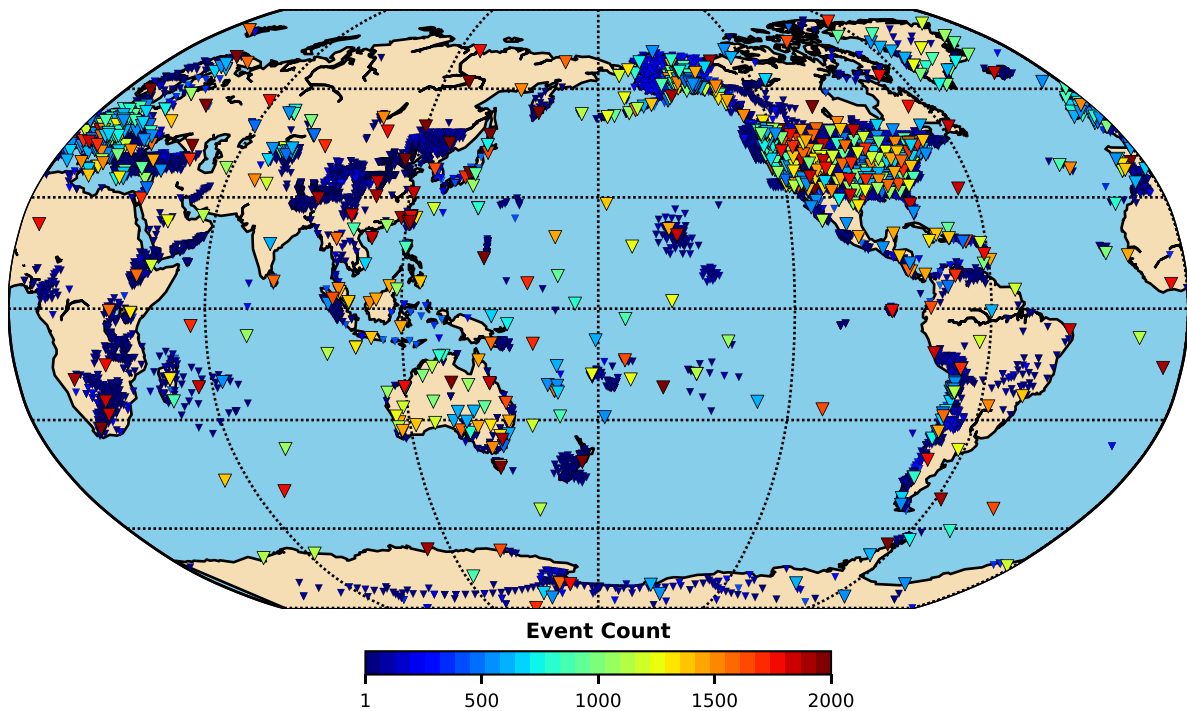


Figure 2. Distribution of the 11 575 seismographic stations used in this study. The colour of a station represents the number of earthquakes in the data set it recorded. (And the winner is: IU MAJO in Matsushiro, Japan, with 2081 earthquakes!) We observe relatively low data coverage underneath the oceans, which may lead to low resolution beneath this area. We mitigate this problem by including more major arc surface waves and body-wave phases that reflect from the ocean floor, including PP, SS, ScSScS, etc.

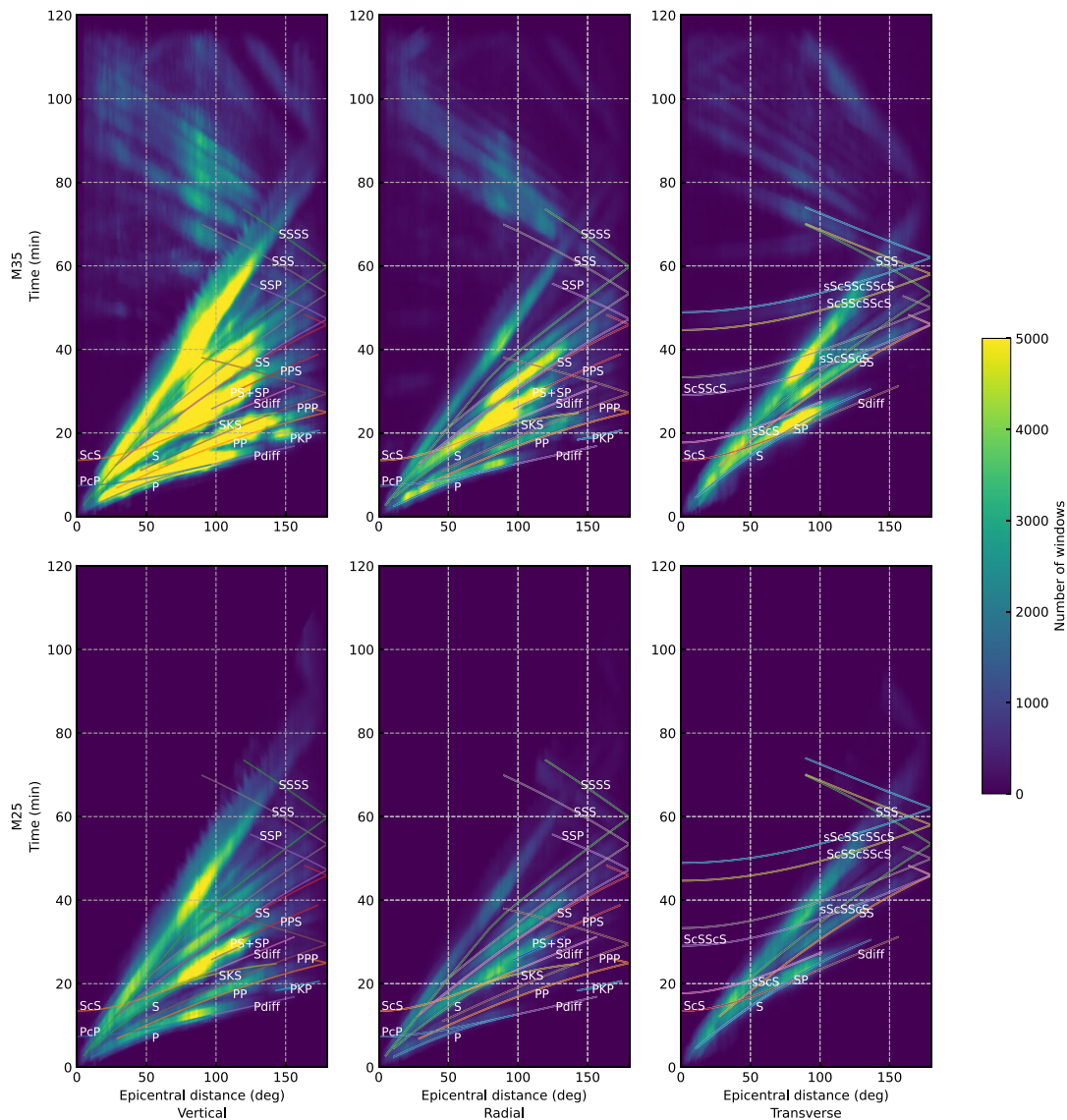


Figure 3. Heatmaps of windows selected in the 17–100 s period band for models GLAD-M35 (top row) and GLAD-M25 (bottom row). The new window selection algorithm used to construct GLAD-M35 is designed to pick more phases before the *S* wave, constraining compressional wave speed structure, and after the surface waves, constraining the deep mantle and structure beneath ocean basins. The total number of GLAD-M35 measurement windows is 40 million, more than double the count for GLAD-M25.

Table 1. Comparison of the data sets used to construct models GLAD-M25 (Lei *et al.* 2020) and GLAD-M35.

	GLAD-M25	GLAD-M35
Number of earthquakes	1480	2160
17–40 s <i>P</i> -wave category	✗	✓
40–100 s <i>P</i> -wave category	✗	✓
17–40 s surface-wave category	✗	✓
Major-arc surface waves	✓	✓
Major-arc body waves	✗	✓
ScS reverberations	✗	✓
Number of measurement windows	18 million	40 million

models with a disjunction between bulk sound speed and shear wave speed, but no significant anticorrelation was revealed. Tesoniero *et al.* (2015) inferred hydration of the mantle through slab-related

processes using the V_p/V_s ratio. Lu *et al.* (2019) refined images of various subducting slabs through joint *P* and *S* inversion.

In the context of adjoint-based tomography, where the propagation of seismic waves is numerically simulated in three dimensions, implementing a joint *P* and *S* inversion is straightforward. By selecting data from time windows corresponding to different phase arrivals (Maggi *et al.* 2009), we can balance *P*- and *S*-wave contributions by individually weighting measurement windows using strategies similar to Ruan *et al.* (2019). However, the relative weighting between *P* and *S* waves is arbitrary, and we observe a systematic long-distance *P*-wave traveltimes error in GLAD-M25 (Lei *et al.* 2020), which used this balancing technique. Rebalancing the measurements for a better compressional wave speed structure is one of the main purposes of this study.

The uncertainty of a tomographic model is an important attribute. In addition to telling us how reliable certain model features are, uncertainty is required in geochemical analysis, such as estimating the

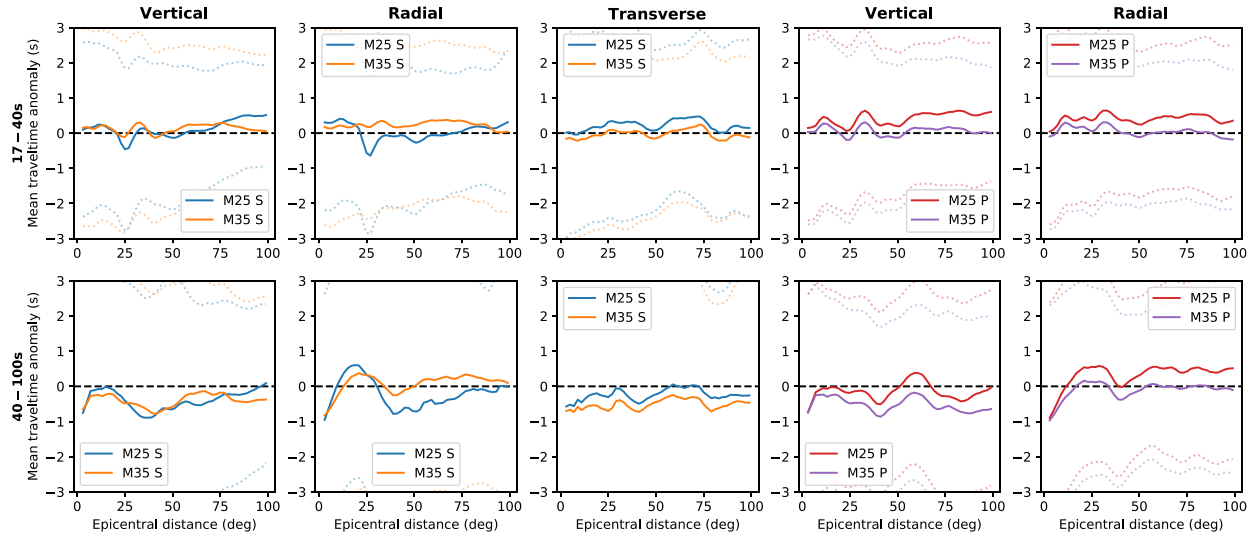


Figure 4. Comparison of GLAD-M25 and GLAD-M35 traveltime anomalies for first-arriving S waves (Columns 1, vertical, 2, radial and 3, transverse) and P waves (Columns 4, vertical, and 5, radial) binned by epicentral distance in the 17–40 s (top row) and 40–100 s (bottom row) period bands. Solid lines denote the mean value in a given epicentral distance bin, and the two dotted lines denote the standard deviation range. For model GLAD-M25, we observe a systematic bias of mean P -wave traveltimes (green curves) at large epicentral distances (> 40 – 60°), which is largely eliminated for model GLAD-M35 (red curves). For S waves, there is no systematic traveltime bias in either model.

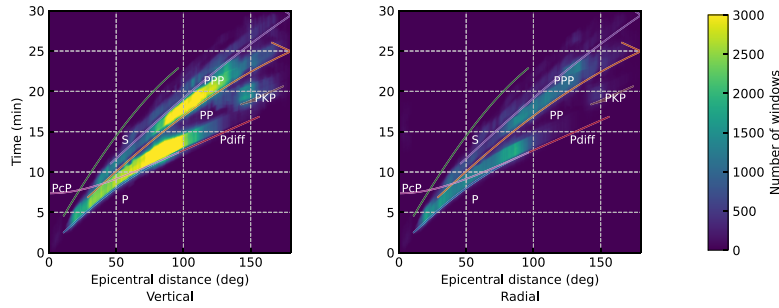


Figure 5. Window selections for P waves in the 17–40 s period band. Left-hand panel: vertical component. Right-hand panel: radial component. Since P arrivals are only observed on the vertical and radial components, no windows are selected on the transverse component.

relative perturbations between model parameters (Ishii & Tromp 2001). It is also useful for future adjoint or Bayesian inversions because uncertainty is needed to utilize prior information. However, quantifying the uncertainty of adjoint tomography models is challenging. Synthetic inversions with oversimplified models, like checkerboard tests, are unreliable (Léveque *et al.* 1993). Based on the assumption of a Gaussian model prior and a linearized approximation of the forward modelling, the posterior model covariance can be estimated by calculating the inverse Hessian of the data misfit (Tarantola 2005). Direct computation of the full Hessian is generally not feasible due to the large number of model parameters, and various attempts have been made to estimate features of the Hessian or its inverse. At a minimum, uncertainty quantification should provide two attributes of the Hessian: the standard deviation, which is directly obtained from the diagonal Hessian terms, and resolution length, which is captured by the off-diagonal Hessian terms. Fichtner & Trampert (2011) used point-spread functions to assess the Hessian of a small local model perturbation. This approach has since been adopted to estimate the resolution length but cannot provide a quantitative estimate of the standard deviation of each model parameter. Liu & Peter (2019) combined the square-root variable metric (SRVM) algorithm with randomized singular-value

decomposition (RSVD) to estimate the inverse Hessian and Liu *et al.* (2022) applied the method based on the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm. This approach can estimate diagonal terms (standard deviations) and off-diagonal terms (resolution length) at a reasonable computation cost. In this paper, we apply this method on a global scale.

The global adjoint tomography (GLAD) project aims to obtain relatively high-resolution images of the Earth's interior by performing adjoint tomography on large-scale HPC clusters. Starting by combining 3-D mantle model S362ANI (Kustowski *et al.* 2008) with 3-D crustal model Crust2.0 (Bassin 2000), the first-generation model, GLAD-M15 (Bozdağ *et al.* 2016), was obtained from fifteen conjugate gradient iterations based on a data set of 253 earthquakes. The first twelve iterations utilized data with a shortest period of 30 s, and the last three included 17 s body-wave data. In contrast, the shortest period of surface waves gradually decreased from 60 to 45 s. The second-generation GLAD-M25 model involved ten additional L-BFGS iterations utilizing 1480 earthquakes. The data set consisted of body waves with periods of 17 s and longer and surface waves with periods of 40 s and longer. Here, we present our third and final generation model, GLAD-M35. On the data side, we made three improvements. First, we added 680 deep or relatively

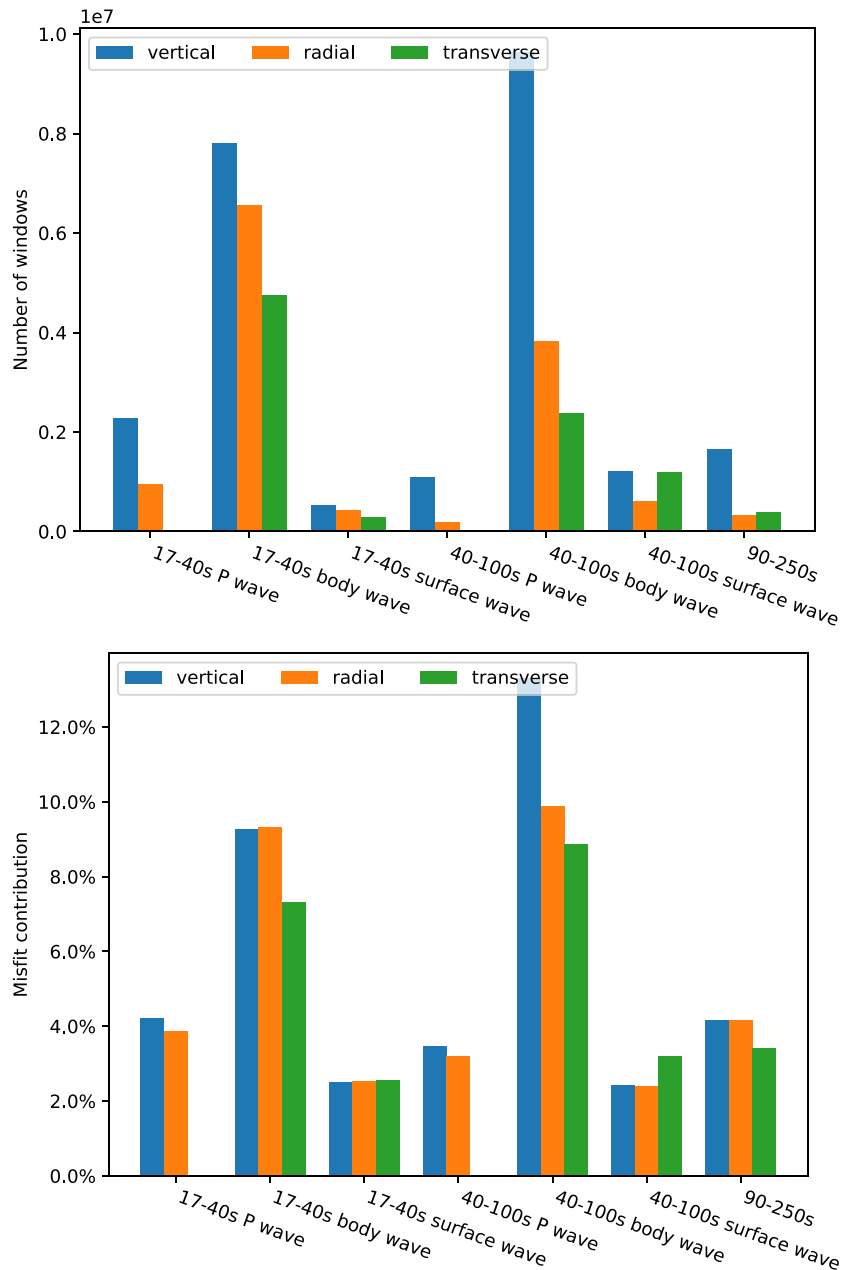


Figure 6. Bar graphs of wave category contributions to window count (top panel) and misfit (bottom panel) based on GLAD-M25. Seven new categories have been added to the inversion: 17–40 s *P* phases on the vertical and radial components; 40–100 s *P* phases on the vertical and radial components; 17–40 s surface waves on the vertical, radial, and transverse components.

recent well-recorded events, for a total of 2160 earthquakes. Secondly, we improved the window selection tool, FLEXWIN (Maggi *et al.* 2009), to include more major-arc phases, thereby more than doubling the number of measurement windows to over 40 million. Thirdly, we incorporated a new *P*-related wave category to better constrain the *P* wave speed. We preserved the model update history and updated the model with BFGS directions. This enabled us to estimate the inverse Hessian with random probing and present an estimation of the uncertainty based on the method introduced by Liu *et al.* (2022). As the final model of the GLAD series, GLAD-M35 features improved constraints on global *P*-wave heterogeneity and the structure of the deep mantle. Together with an estimation of

uncertainty, it can be used for a wide range of geodynamic and geochemical analyses and serve as the starting model of future adjoint or Bayesian inversions.

This paper is organized as follows. Section 2 introduces the data and new measurements. Section 3 introduces the workflow of the inversion, which includes forward/adjoint simulations (Section 3.1) and the selection of a misfit function (Section 3.2). Section 4 discusses the results, including the misfit evolution (Section 4.1), analysis of the final model GLAD-M35 and comparison with other models (Section 4.2), and highlights the importance of the new *P* category. Section 5 discusses the uncertainty and resolution estimation of the model.

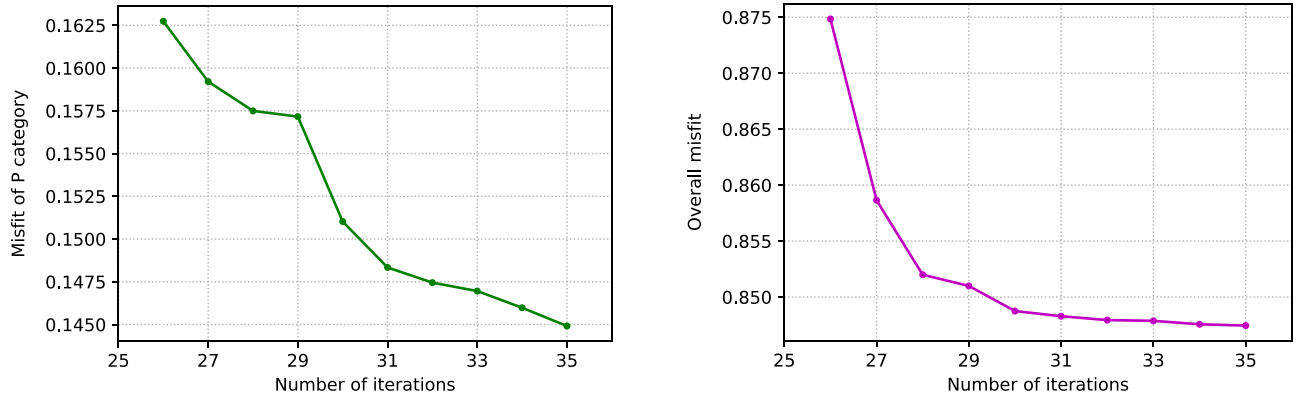


Figure 7. Evolution of the P -wave category misfit (left-hand panel) and the overall multi-taper phase misfit (right-hand panel) as a function of the number of iterations from GLAD-M25 to GLAD-M35.

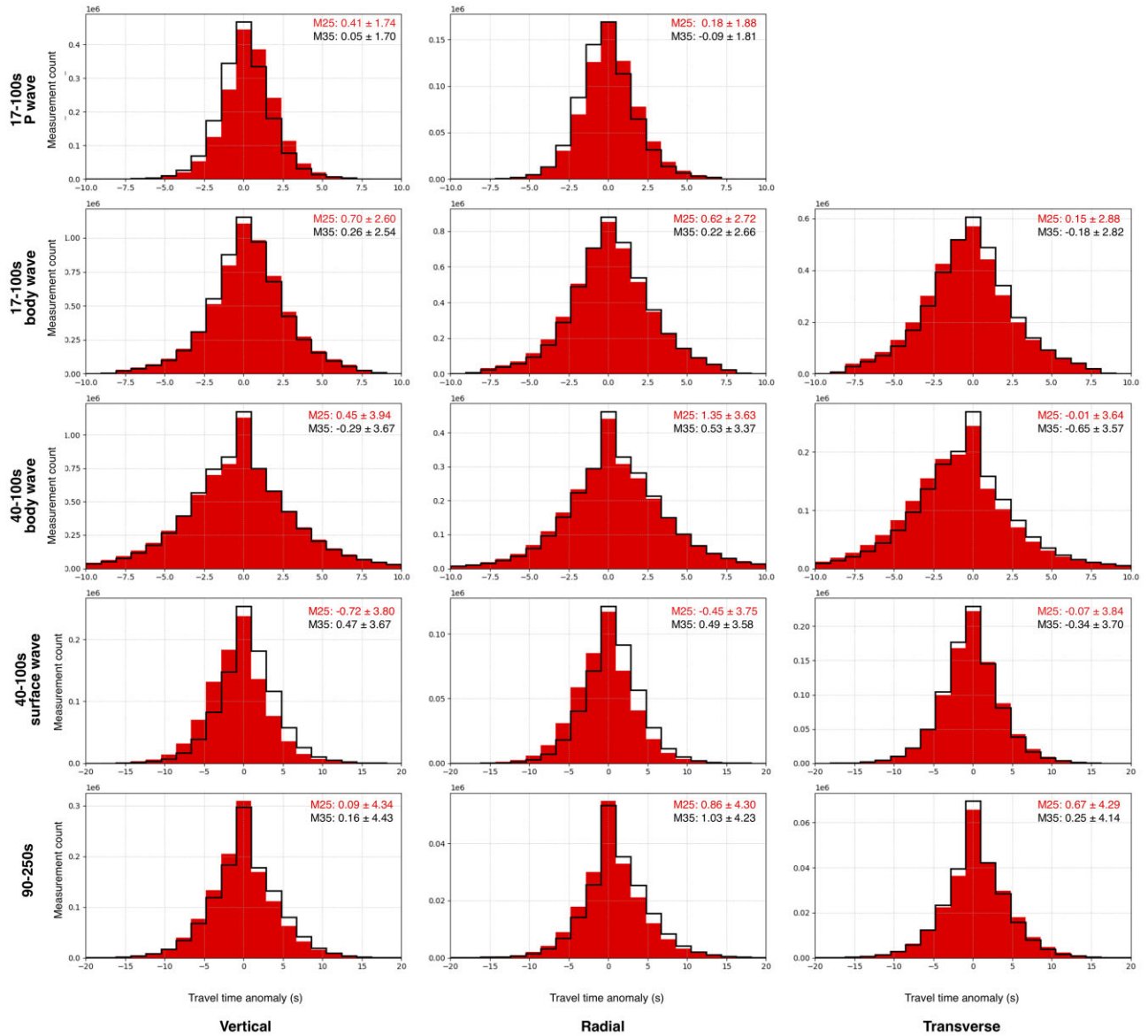


Figure 8. Traveltime anomaly histograms for GLAD-M25 and GLAD-M35. Red bins show the statistics for GLAD-M25, and black lines show the statistics for GLAD-M35. From left to right, columns correspond to the vertical, radial, and transverse components, and rows from top to bottom correspond to 17–40 s P waves, 17–40 s body waves, 40–100 s body waves, 40–100 s surface waves and 90–250 s surface waves. Overall, the histograms for GLAD-M35 are more centred and narrower compared to GLAD-M25, and we observe that the P -wave histograms are more centred at zero in GLAD-M35, in agreement with Fig. 4.

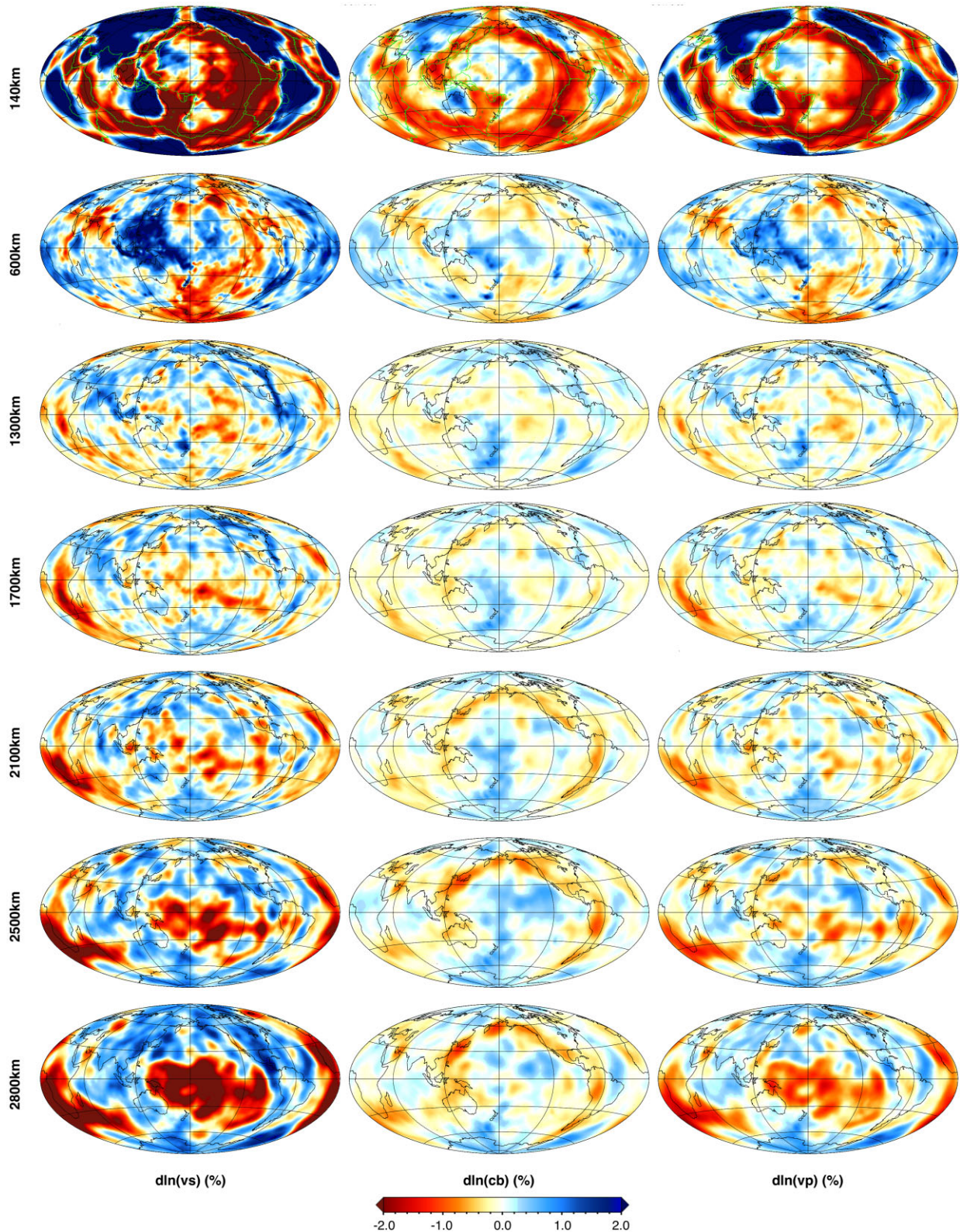


Figure 9. Map views of model GLAD-M35 at depths of 140 km, 600 km, 1300 km, 1700 km, 2100 km, 2500 km and 2800 km, showing shear wave speed perturbations (left-hand column), bulk sound speed perturbations (middle column) and compressional wave speed perturbations (right-hand column).

2 DATA AND INITIAL MODEL

We use GLAD-M25 as our initial model (see Appendix A). The model parameters are based on the five Love parameters, A , C , L , N and F , with the assumption that radial anisotropy is due

to shear anisotropy. Thus, four independent parameters are included in the inversion: bulk sound speed $c = \sqrt{(C - 4L/3)/\rho} = \sqrt{(A - 4N/3)/\rho} \equiv \sqrt{\kappa/\rho}$, which is defined in terms of the bulk modulus κ and the mass density ρ , the shear wave speeds of

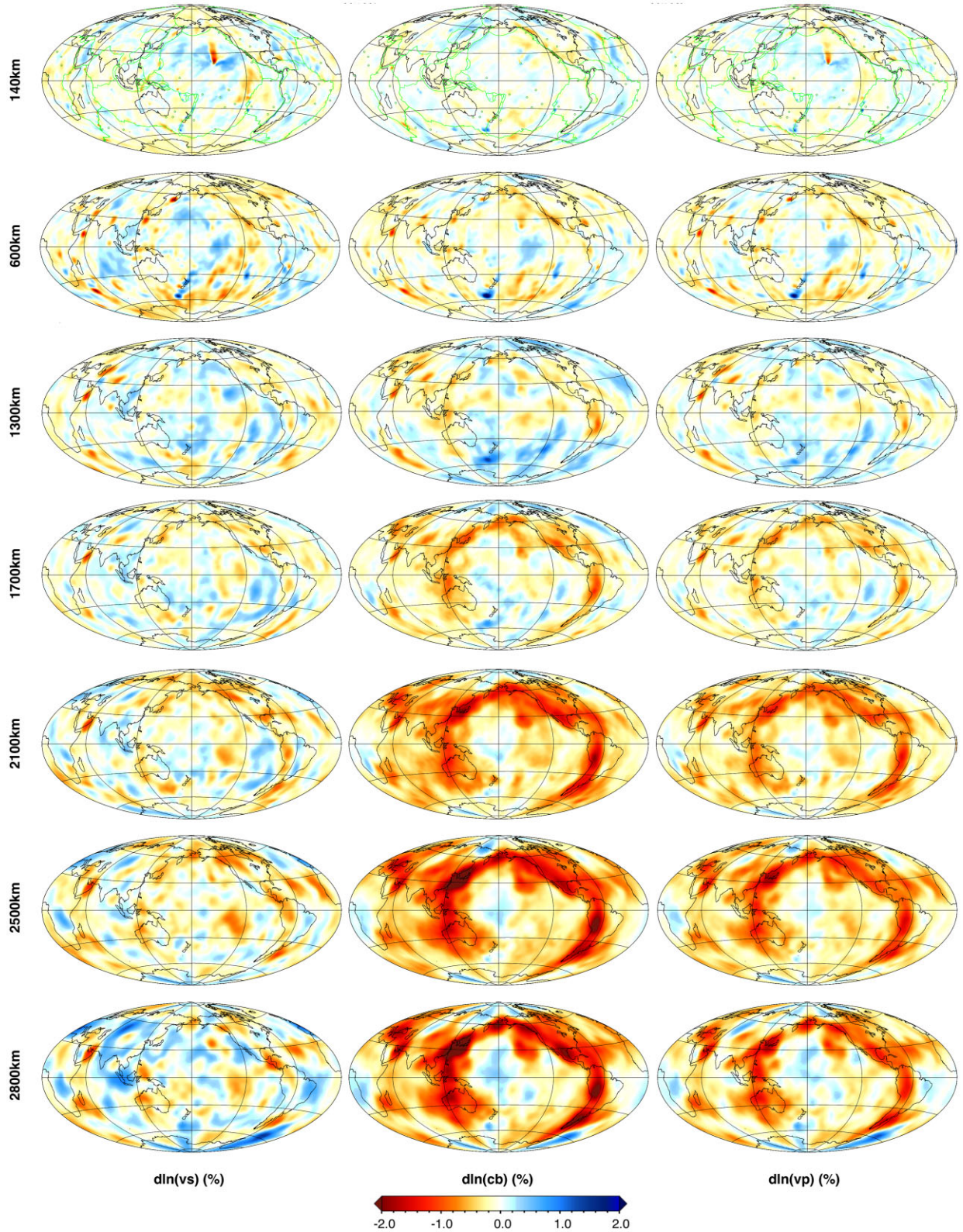


Figure 10. Model updates of GLAD-M35 relative to GLAD-M25 at depths of 140 km, 600 km, 1300 km, 1700 km, 2100 km, 2500 km and 2800 km showing shear wave speed perturbations (left-hand column), bulk sound speed perturbations (middle column) and compressional wave speed perturbations (right-hand column). The most significant changes are in the bulk sound speed at depths greater than 2000 km.

horizontally and vertically polarized shear waves $\beta_h = \sqrt{N/\rho}$ and $\beta_v = \sqrt{L/\rho}$, and the dimensionless parameter $\eta = F/(A - 2L)$. A total of 1480 earthquakes were used to construct GLAD-M25.

The measurements were selected in four categories: 17–40 s body waves, 40–90 s body waves, 40–90 s surface waves and 90–250 s surface waves.

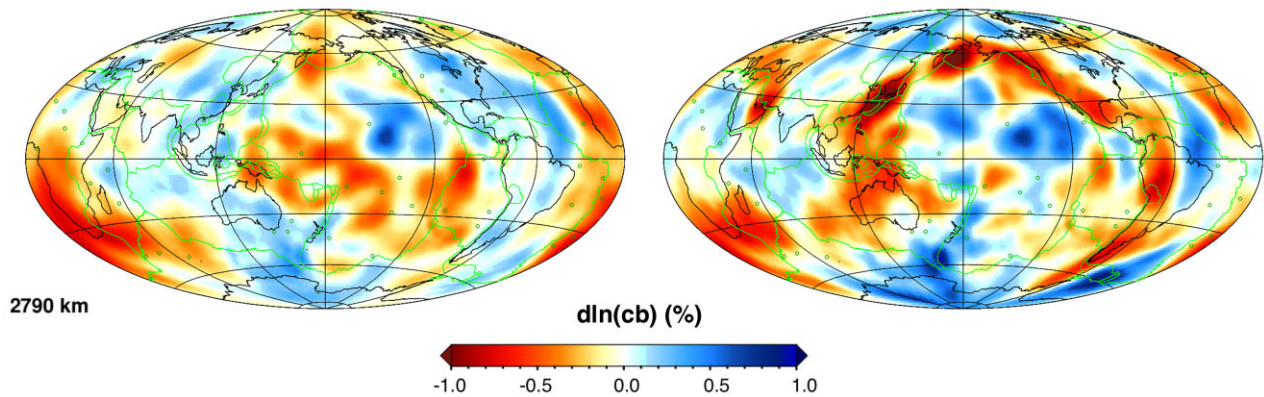


Figure 11. Comparison between bulk sound speed perturbations in GLAD-M25 (left-hand panel) and GLAD-M35 (right-hand panel) at a depth of 2790 km. GLAD-M35 has a distinct circum-Pacific imprint.

For the construction of model GLAD-M35, we added 680 earthquakes to our database, emphasizing deep earthquakes (>300 km) before 2015 and well-recorded earthquakes after 2015. Fig. 1 illustrates the overall earthquake database. Fig. 2 shows the distribution of the 11 575 seismographic stations used in the inversion, coloured by the number of earthquakes they recorded.

On top of the four measurement categories used for the construction of GLAD-M25, we added several new wave categories: 17–40 s surface waves, 17–40 s *P*-related phases, and 40–100 s *P*-related phases, which we detail in Section 2.1. We made two major improvements to our window selection algorithm. First, we fine-tuned the algorithm to pick ScS and its multiples. Second, we tuned the algorithm to pick majorarc body- and surface-wave phases after 60 min. Fig. 3 shows 17–100 s heatmaps of window selections based on the new selection criteria for GLAD-M35, which are contrasted with the corresponding GLAD-M25 heatmaps. On the transverse component, we observe that up to ScS4 windows are picked by the algorithm. In the body-wave categories, we see that more body phases are selected after the minor-arc surface waves, and in the surface-wave categories, we see that more major-arc surface waves were selected. This increases ray coverage in the deep mantle and underneath ocean basins.

For our previous earthquake database, we were able to pick 18 million windows. Based on the new earthquake database and the improved window-picking algorithm, we selected over 40 million windows. On average, for each earthquake, we pick 50 per cent more windows. Table 1 summarizes the differences between the GLAD-M25 and GLAD-M35 data sets.

2.1 *P*-wave categories

A thorough investigation of *P* and *S* traveltime anomalies (Fig. 4) reveals a systematic bias for compressional wave speed at long epicentral distances, whereas no significant bias is found in shear wave speed. This is partly because the initial model of the GLAD project, S362ANI, does not have good constraints on compressional wave speed. Another major cause of this problem is that the number of *P*-related phases is much smaller than that of *S*-related phases. *P*-related phases only contribute ~4 million measurement windows out of 40 million measurements (10 per cent). In the GLAD-M25 weighting strategy, *S*-related phases dominate the misfit function, and the inversion prioritizes improvements in the shear-wave

structure. To overcome this problem, we added four new *P*-wave categories to the misfit function, including (1) 17–40 s vertical component *P*-related phases; (2) 17–40 s radial component *P*-related phases; (3) 40–100 s vertical component *P*-related phases and (4) 40–100 s radial component *P*-related phases. The newly added categories focus on assimilating phases directly sensitive to the Earth's compressional wave speed structure.

Fig. 5 shows window selection results for *P*-related arrivals in the 17–40 s period band. On the vertical component (left-hand panel), there are multiple *P*-related selections, including *P*, *P*diff, *PP* and *PKP*, with similar results for the radial component (right-hand panel). No windows are selected on the transverse component, since most *P*-related phases are only observed on the vertical and radial components.

A benefit of creating new *P*-wave-specific categories is that we can assign dedicated category weightings to such phases to increase their contributions to the overall misfit function. Fig. 6 shows relative contributions to window counts and misfits from each measurement category. A good example is the 17–40 s *P*-related phase on the radial component. It contributes 2.1 per cent relative to the overall number of windows. However, through our category weightings, it contributes 3.9 per cent to the overall misfit function.

3 ADJOINT TOMOGRAPHY WORKFLOW

The inversion is performed on the ‘Summit’ cluster at OLCF, which uses IBM Power9 CPUs and NVIDIA Tesla V100 GPUs. We adopted the same workflow as is described in Lei *et al.* (2020). We downloaded observed data from IRIS (iris.edu) and ORFEUS (orfeus-eu.org), followed by data processing and window selection. Each iteration involves five main steps for each event: (1) forward simulations to obtain synthetic seismograms, (2) calculation of the misfit function, (3) adjoint simulations and kernel calculations, (4) kernel processing to determine the model update search direction and (5) line search to determine the step length.

3.1 Forward/adjoint simulations

We run forward and adjoint simulation using the spectral-element solver SPECFEM3D_GLOBE (Komatitsch & Tromp 2002b, a). The simulation time is 120 min with a shortest period of 17 s. For each event, 384 MPI processes are used, which requires 96 GPUs with the

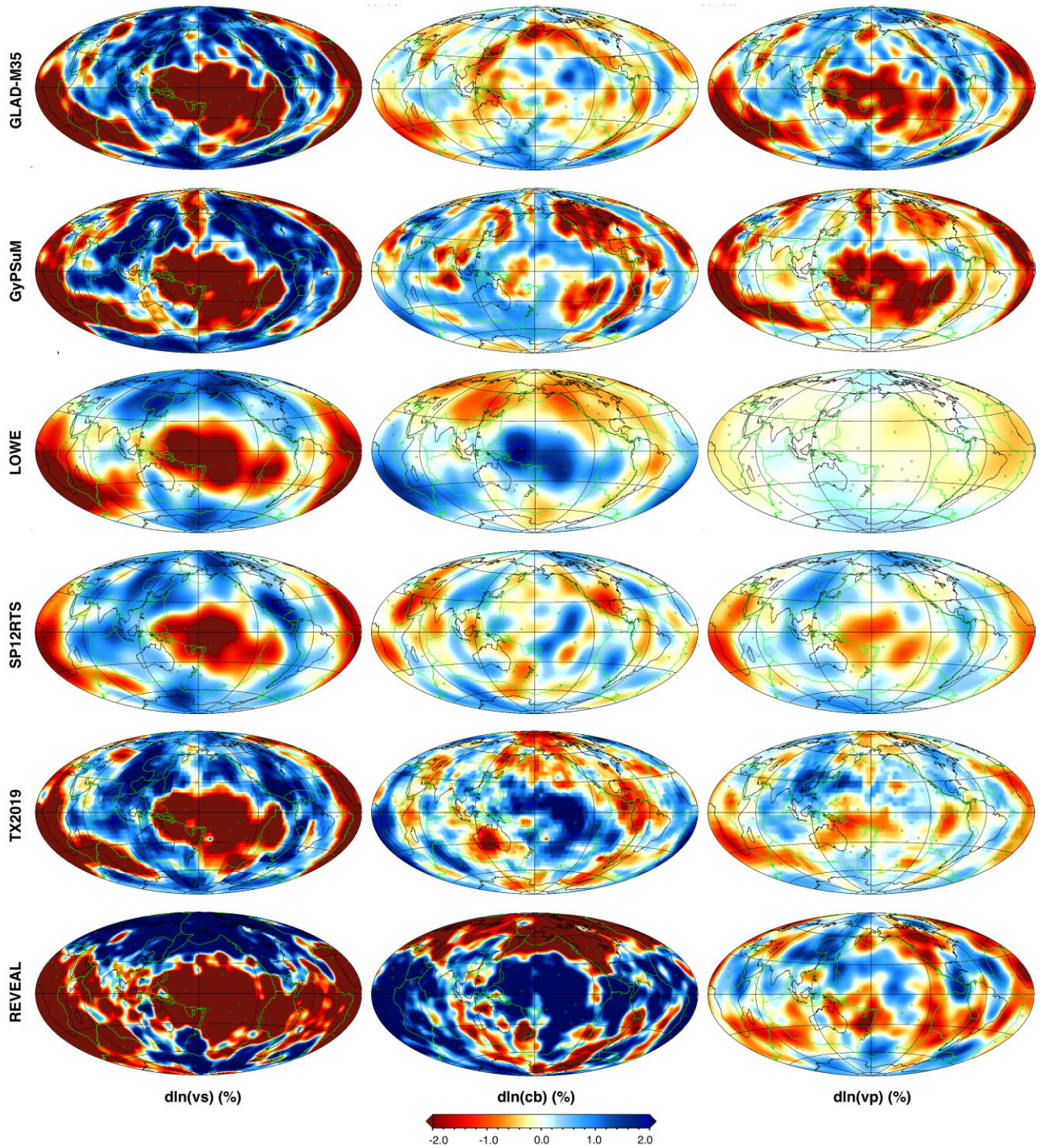


Figure 12. Joint P and S models at a depth of 2800 km, plotted relative to the 1-D mean of the REVEAL and LOWE models and the 1-D reference model of each of the other models. (We use the same plotting convention for all model images.) First row: GLAD-M35. Second row: GyPSuM (Simmons *et al.* 2010). Third row: LOWE (van Herwaarden *et al.* 2020). Fourth row: SP12RTS (Koelemeijer *et al.* 2016). Fifth row: TX2019 (Lu *et al.* 2019). Sixth row: REVEAL (Thrustarson *et al.* 2024). The columns show shear wave speed perturbations (left-hand panel), bulk sound speed perturbations (middle panel) and compressional wave speed perturbations (right-hand panel).

Multi-Process Service (MPS) technology. This means that 16 nodes are required for each event. We run the 2160 events with 4 jobs and run 20 events at a time in each job. This means that each job requires 320 nodes, and the wall time is 27 times the cost of a single event simulation.

A total of 34 snapshots of the forward wavefield are saved for kernel computation. ADIOS binary pack format (Liu *et al.* 2014), which is well-optimized for parallel writing, is used to store large chunks of data, including models, kernels, mesh files and wavefield snapshots.

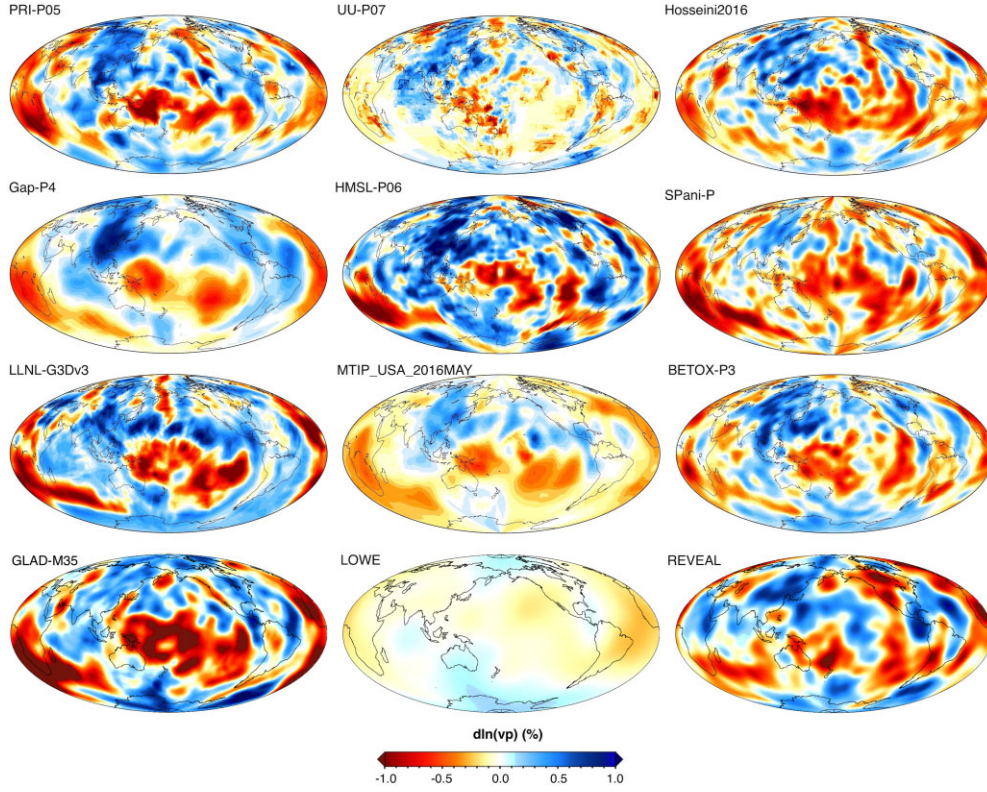


Figure 13. Twelve P models at a depth of 2790 km. From the top left to the bottom right, shown are PRI-P05 (Montelli *et al.* 2006), UU-P07 (Rocha *et al.* 2019), Hosseini2016 (Hosseini 2016), GAP-P4 (Obayashi *et al.* 2013), HMSL-P06 (Houser *et al.* 2008), SPani-P (Tesoniero *et al.* 2015), LLNL-G3Dv3 (Simmons *et al.* 2012), MITP_USA_2016MAY (Burdick *et al.* 2008), DETOX-P3 (Hosseini *et al.* 2020), M35, LOWE (van Herwaarden *et al.* 2020), REVEAL (Thrustarson *et al.* 2024). The first nine models of the figure were plotted with the SubMachine web-based tool (Hosseini *et al.* 2018). Although some models agree at longer wavelengths, there are major differences at smaller length scales.

3.2 Misfit function

Following Lei *et al.* (2020), the misfit function for model $\mathbf{m}$ is defined as

$$\Phi(\mathbf{m}) \equiv \sum_{s=1}^S w_s \sum_{c=1}^C w_c \sum_{r=1}^{R_{sc}} w_{scr} \sum_{w=1}^{W_{scr}} \chi_{scrw}(\mathbf{m}) + (\mathbf{m} - \mathbf{m}_0)^t \mathbf{C}_m^{-1} (\mathbf{m} - \mathbf{m}_0), \quad (1)$$

where S denotes the number of sources, C the number of categories, R_{sc} the number of receivers recording source s in category c , and W_{scr} the number of measurement windows for source s , category c and receiver r . The second term on the right-hand side of eq. (1) serves to limit the differences between the current model $\mathbf{m}$ and the initial model $\mathbf{m}_0$ based on the prior model covariance $\mathbf{C}_m$; a superscript t denotes the transpose. If such a prior is not known, it can be replaced by $\varepsilon \mathbf{I}$, where $\mathbf{I}$ denotes the identity matrix and ε a suitably chosen parameter to balance the two terms in eq. (1).

Windows in the same seismogram are given a uniform weight. Multitaper phase misfit (e.g. Laske & Masters 1996; Zhou *et al.* 2004) is used as the measurement, resorting to cross-correlation measurements when the window is too short. For a specific source s , category c , receiver r and window w , the misfit is defined as

$$\chi_{scrw}(\mathbf{m}) = \frac{1}{\Delta\omega} \int_{\omega_1}^{\omega_2} \left(\frac{\Delta\tau_{scrw}(\omega; \mathbf{m})}{\sigma_{scrw}} \right)^2 d\omega, \quad (2)$$

where $\Delta\tau_{scrw}(\omega; \mathbf{m})$ denotes a multitaper frequency-dependent phase measurement over the frequency interval $\Delta\omega = \omega_2 - \omega_1$,

with associated uncertainties σ_{scrw} . Event weights, w_s , and station weights, w_{scr} , are determined by the geographical distribution of the events and stations of an event in a specific category (Ruan *et al.* 2019). The event weights are normalized so that

$$\sum_{s=1}^S w_s = S. \quad (3)$$

For event s in category c , the station weights are normalized by the number of measurements in the category:

$$\sum_{r=1}^{R_{sc}} w_{scr} W_{scr} = R_{sc}. \quad (4)$$

Finally, category weights are chosen to make sure each category has an even contribution to the overall misfit:

$$w_c \sum_{s=1}^S w_s \sum_{r=1}^{R_{sc}} w_{scr} W_{scr} = w_c \sum_{s=1}^S w_s R_{sc} = \frac{1}{C}. \quad (5)$$

The contributions of the categories to the misfit function are illustrated in Fig. 6(right-hand panel).

4 RESULTS

4.1 Misfit evolution

Fig. 7 shows the P -wave category misfit (left-hand panel) and the overall multitaper phase misfit (right-hand panel) as a function of the number of iterations from GLAD-M25 to GLAD-M35. We performed two BFGS inversions, each containing five iterations.

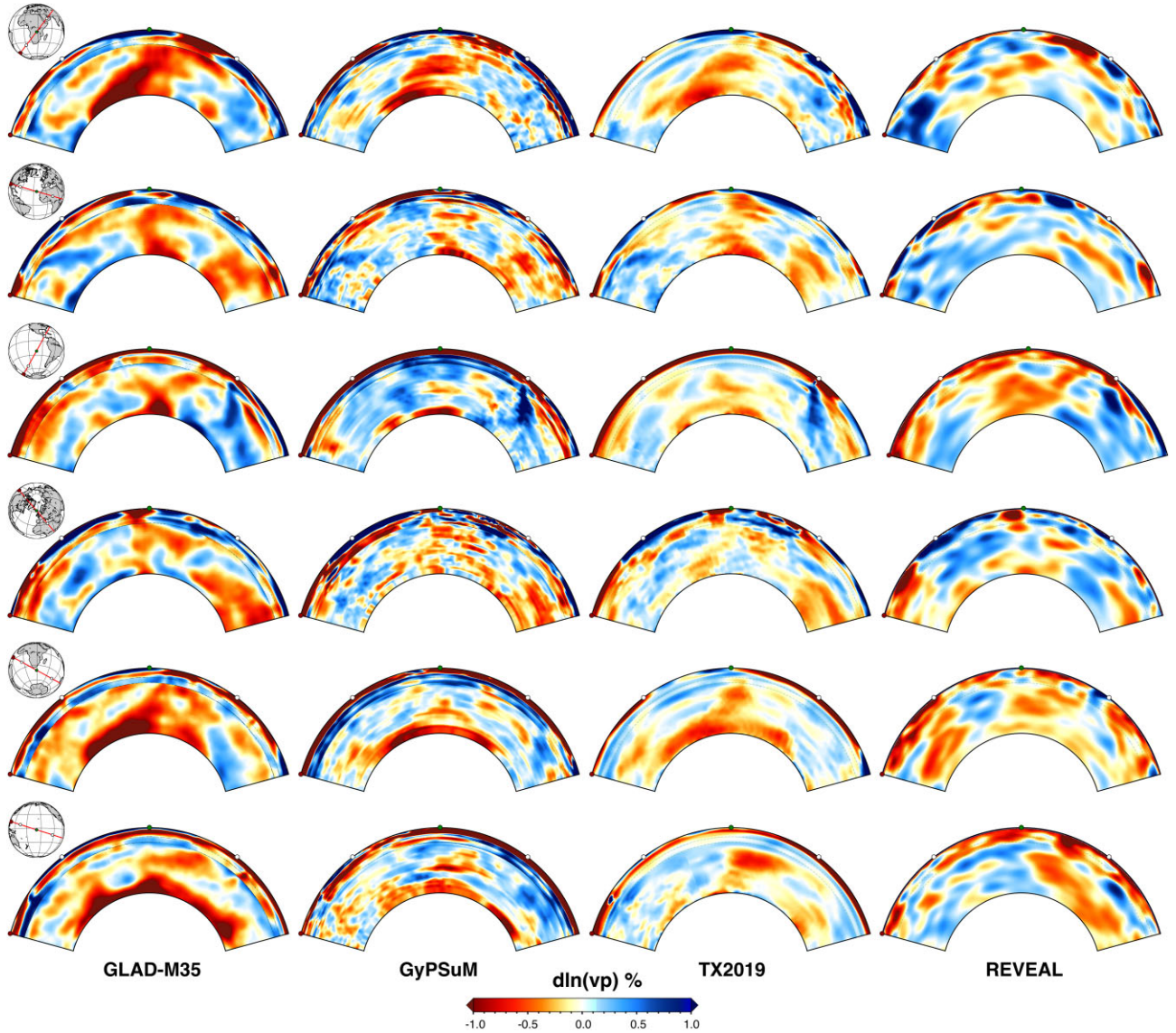


Figure 14. Vertical cross-sections of *P* wave speed in models GLAD-M35 (Column 1), GypSuM (Simmons *et al.* 2010) (Column 2), TX2019 (Lu *et al.* 2019) (Column 3) and REVEAL (Thrustarson *et al.* 2024) (Column 4). The small maps in the top left of each row show map views of the cross-section in that row.

The same data set was used in both inversions, the difference being that the first inversion has a smoothing length of 100 km, whereas the second uses a smoothing length of 75 km. We observe a steady decrease in both misfits, and the fact that the overall misfit converges justifies performing a linearized uncertainty analysis.

As illustrated in Fig. 4, the systematic *P*-wave traveltime anomaly in GLAD-M25 has been largely eliminated in GLAD-M35. Fig. 8 shows traveltime anomaly histograms for GLAD-M25 and GLAD-M35 in each misfit category. We observe a significant misfit decrease in 17–40 s *P* waves, 17–40 s body waves, and 40–100 s body waves, and the distributions now almost centre at zero. The reason the misfit in GLAD-M25 is not centred at zero originates in the initial model of the GLAD project, S362ANI, which is a shear-wave model utilizing only low frequencies. By comparing the result with Figs 10 and 11 in Lei *et al.* (2020), we see that throughout all 35 iterations of the GLAD project, the centre of the distributions slowly moves toward zero. However, we do notice that not all categories have consistent misfit reductions, and we attribute this to the limited degrees of freedom in the model parametrization. For example,

introducing azimuthal anisotropy should further decrease the misfit in the surface-wave categories.

4.2 GLAD-M35

4.2.1 Comparison with model GLAD-M25

In this section, we compare models GLAD-M25 and GLAD-M35, highlighting the new contributions of the latter. Fig. 9 shows depth slices of GLAD-M35, and differences between GLAD-M35 and GLAD-M25 are illustrated in Fig. 10. In all maps and cross-sections, the one-dimensional average model (see appendix A of Lei *et al.* 2020), summarized in Fig. A2 in Appendix B, has been removed. While there are no significant changes between GLAD-M25 and GLAD-M35 in the upper mantle, except for the Tonga subduction zone and Hawaii, the update in the lower mantle is more pronounced. This is thanks to the new window selection criteria that include many more body-wave phases after the arrival of the minor-arc surface waves. We observe that the new *P*-wave categories have

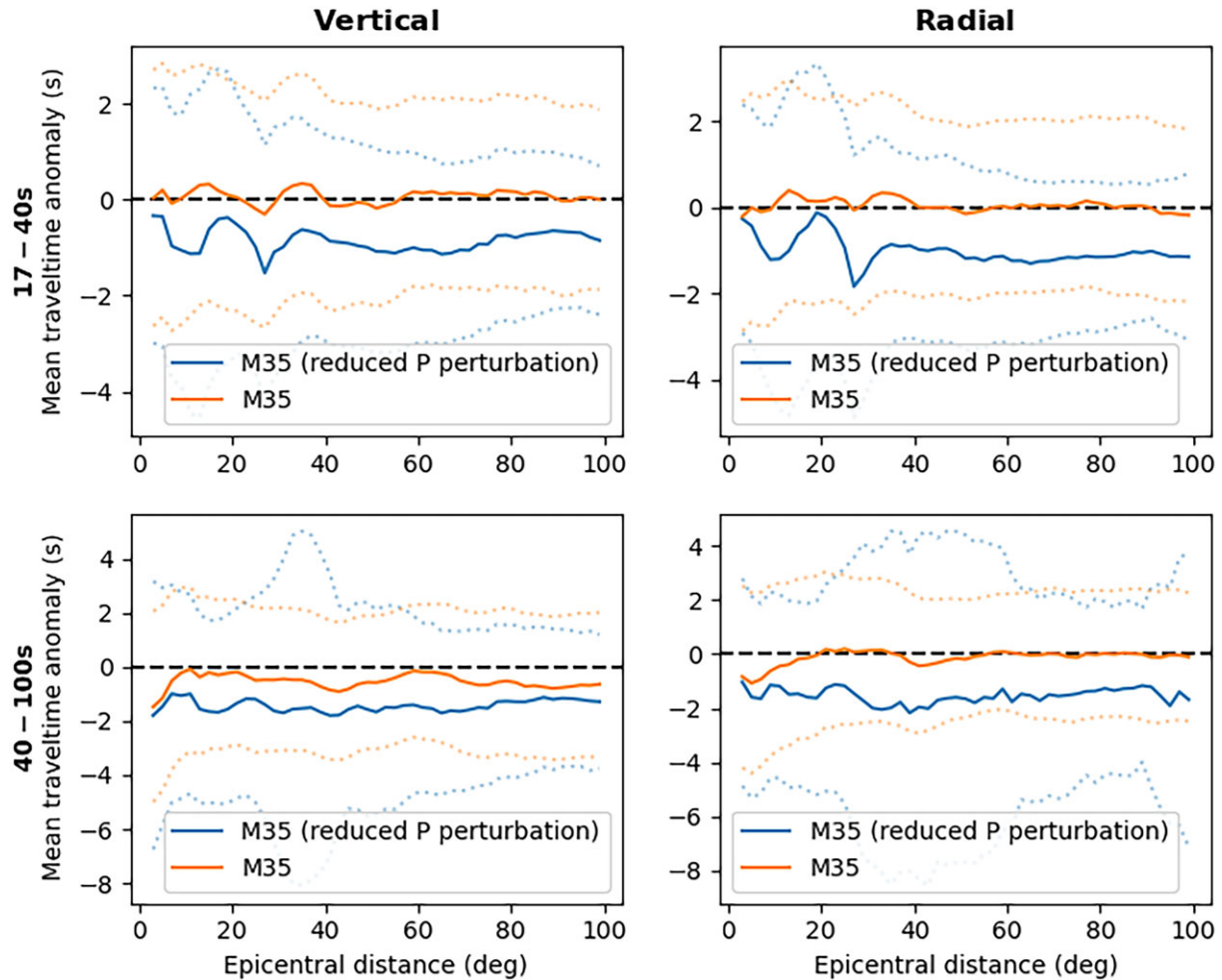


Figure 15. *P*-wave traveltime predictions based on GLAD-M35 (Orange lines) contrasted with predictions based on GLAD-M35 with a 50 per cent reduction in the *P* wave speed perturbations (blue lines). We observe that reducing the *P* wave speed perturbations causes the mean traveltime to deviate substantially from zero, thereby illustrating that the *P* wave speed perturbations in GLAD-M35 have the correct order of magnitude.

a more significant impact on *P*-wave speed updates than *S*-wave speed updates. This compensates for the lack of weight on the *P* waves in the GLAD-M25 misfit function.

Model GLAD-M35 reflects the new window selection criteria and the *P*-phase categories, indicating the effective selection of new phases and a good fit for the original GLAD-M25 data set. While we have a significant model update in the lower mantle, the overall difference between GLAD-M35 and GLAD-M25 is much smaller than between GLAD-M25 and GLAD-M15. This is further evidence that the inversion is very close to convergence and that no significant model updates can be made unless we increase the number of events, use higher frequencies, or introduce new model parameters.

Upon comparing the bulk sound speed perturbations between GLAD-M25 and GLAD-M35 at a depth of 2790 km (Fig. 11), we observe a change in sign beneath various locations, including East Asia, the South Pacific, and Central America. The bulk sound speed in GLAD-M25 is correlated with the *P* wave speed, a property most joint *P* and *S* models do not exhibit in this region. This suggests that the *P* wave speed in GLAD-M25 has not undergone enough updates to eliminate the imprint of the initial model, S362ANI, in which *P* wave speed is a derived parameter based on a constant Poisson's ratio. This may also explain the systematic error in the traveltime

of the long-distance *P* waves in GLAD-M25. By comparison, bulk sound speed in GLAD-M35 does not have a significant correlation with *P* wavespeed and matches well with various geodynamic constraints described in Simmons *et al.* (2010).

4.2.2 Comparisons with other tomographic models

Fig. 12 shows various joint *P* and *S* models at a depth of 2800 km. There is a striking similarity between GLAD-M35 and GyP-SuM (Simmons *et al.* 2010), which is a model obtained from body-wave traveltimes combined with geochemical data. All models generally agree on the shape of the *P* and *S* perturbations, the main difference being the strength of the *P*-wave anomaly and the geometry of the bulk sound speed perturbations. The anti-correlation between bulk sound and *S* wave speed has compositional and thermal implications (e.g. Masters *et al.* 2000; Oganov & Ono 2004; Koppers *et al.* 2021). Of these models, LOWE (van Herwaarden *et al.* 2020) and TX2019 (Lu *et al.* 2019) have strong global anti-correlations near the core–mantle boundary and relatively weak *P*-wave perturbations. GLAD-M35, GyPSuM and SP12RTS (Koelemeijer *et al.* 2016) only show local anticorrelations and more-or-less agree on the geometry of the bulk sound speed perturbations. As exemplified by the LOWE model, when *P* wave speed perturbations are small

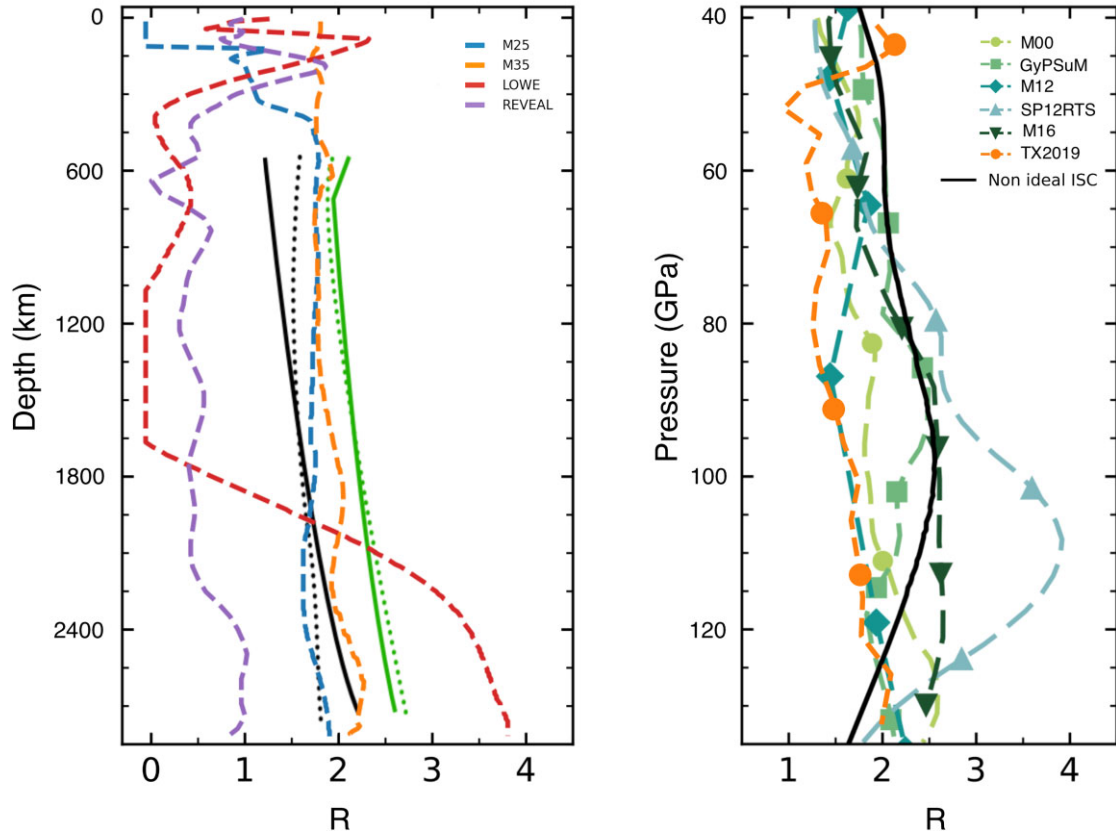


Figure 16. Left-hand panel: S -to- P wave speed perturbation ratio R defined by eq. (6) for models GLAD-M25, GLAD-M35 and LOWE (Thrustarson *et al.* 2022) and REVEAL (Thrustarson *et al.* 2024), and mineral physics estimates from Karato & Karki (2001). Black lines are R values for MgO (solid) and perovskite (dotted), respectively, that only include the effects of anharmonicity. Green lines are R values for MgO (solid) and perovskite (dotted) including the effects of anharmonicity and anelasticity Karato & Karki (2001, Plate 2.a). The smaller than one R values in the lower mantle for the LOWE and REVEAL models are incompatible with mineral physics expectations, nor with the other joint P and S tomographic models shown to the right. Right-hand panel: S -to- P wave speed perturbation ratio for M00 (Masters *et al.* 2000), GyPSuM (Simmons *et al.* 2010), M12 (Mosca *et al.* 2012), M16 (Moulik & Ekström 2016), SP12RTS (Koelemeijer *et al.* 2016), TX2019 (Lu *et al.* 2019) and predicted values based on a non-ideal iron spin crossover (ISC) model (Zhuang & Wentzcovitch 2024).

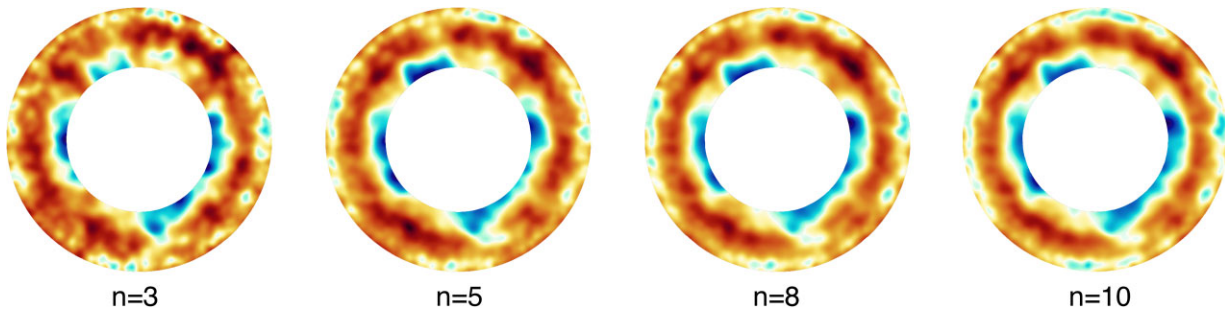


Figure 17. Diagonal values of the inverse Hessian ($(\mathbf{V}\mathbf{A}\mathbf{V}^T + \mathbf{B}_0)$ in eq. 8) probed with $n = 3, 5, 8, 10$ random vectors. The main features of the diagonal Hessian gradually stabilize around $n = 8$.

compared to S wave speed perturbations, there will be a natural anticorrelation between bulk sound speed and S wave speed. Many factors can result in an inaccurate amplitude of the P perturbations, including measurement uncertainty, an imbalance between P and S data in the joint inversion, and a lack of preconditioning. Determining the correct magnitude of the P perturbations is thus the key to the bulk sound speed model.

Fig. 13 shows nine other P models at 2790 km depth. Most of the models agree on the overall geometry of the anomalies, but

we see strong disagreement on the magnitude of the perturbations. Fig. 14 shows P wave speed vertical cross-sections containing various plume systems in GLAD-M35, GyPSuM and TX2019. Models GLAD-M35 and TX2019 have distinct plume features that match well and have similar magnitudes. As an additional line of evidence for large P perturbations, as suggested by GLAD-M35, we ran forward simulations for all 2,160 earthquakes using the GLAD-M35 model with its P wave speed perturbations reduced by half. As shown in Fig. 15, we calculated the same statistics of first-arrival

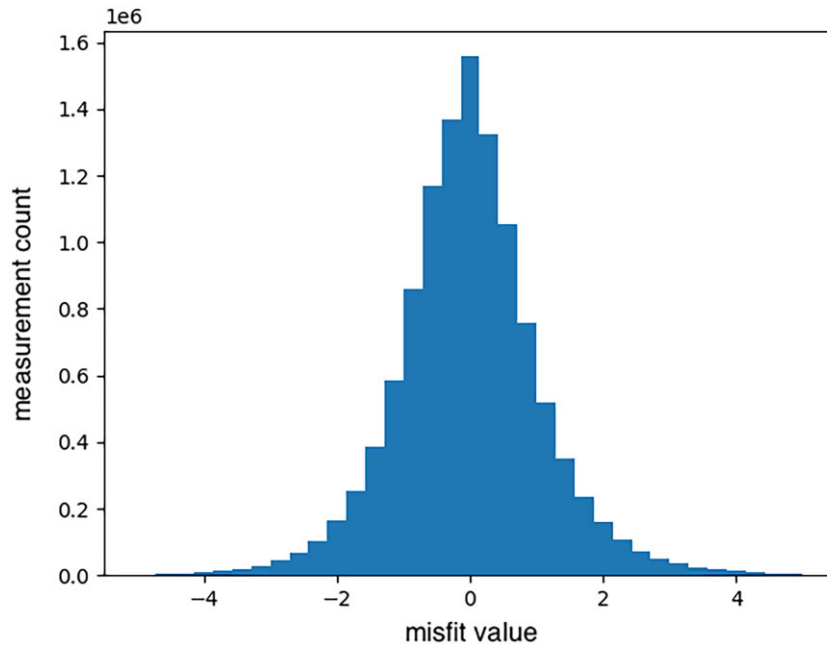


Figure 18. The distribution of multitaper phase misfit values has a standard deviation of 1.09 and, therefore, closely resembles a standard normal distribution, indicating reliable measurement uncertainty estimates.

P waves as in Fig. 4. We see a noticeable offset of the mean traveltime anomaly from zero. The mean traveltime anomaly is never part of the objective function, but it does have implications for the large-scale features of the model. We conclude that reducing the magnitude of the P wave speed perturbations will result in a worse data fit.

4.2.3 S -to- P wave speed perturbation ratio (R value)

Another way to examine the P structure is through the shear-to-compressional (S -to- P) wave speed perturbation ratio, which is given by (for details, see Appendix C)

$$R \equiv \frac{\delta \log V_s}{\delta \log V_p}. \quad (6)$$

A reasonable P wave speed structure is essential to this estimation, which provides useful insight into the mineralogy of the mantle. In general, we would expect the R value to be close to certain known compositions, such as MgO and perovskite. Due to the nature of our least-squares approach, if the R value of a tomographic model is too small, this may be due to an imbalance or a lack of correlation between P and S wavespeed perturbations.

Fig. 16(left-hand panel) shows the R values for models GLAD-M25 and GLAD-M35, as well as the LOWE model (van Herwaarden *et al.* 2020) and its successor REVEAL (Thrustarson *et al.* 2024); the latter are two other joint P and S global models based on adjoint tomography. Fig. 16(right-hand panel) reproduces parts of fig. 4(d) of Zhuang & Wentzcovitch (2024) showing R values for various tomographic models based on body waves and normal modes as well as predicted values for a non-ideal iron spin crossover model. We observe that the R profiles for GLAD-M25 and GLAD-M35 are compatible with mineral physics expectations (Karato & Karki 2001) and other tomographic models, but the LOWE model (involving 72 tomographic iterations) and the REVEAL model (involving 305 iterations) have highly unrealistic S -to- P wave speed perturbation ratios. The R value of the LOWE model is zero in a wide depth

range of around 1500 km, which indicates that P heterogeneity is completely uncorrelated with S heterogeneity. Below ~ 1700 km, the R value increases rapidly to 4, at a rate far beyond any mineral physics expectations. The more recent REVEAL model has an R value that oscillates below one, which is difficult to explain mineralogically. These observations illustrate that while, in theory, using more iterations can compensate for imbalanced model parameters, this brute-force strategy is not guaranteed to work out. Looking at the data to determine the features that best constrain poorly resolved parameters and regions can be crucial in global inversions.

5 UNCERTAINTY QUANTIFICATION

To quantify uncertainty, we follow a similar strategy as described in Liu *et al.* (2022) with a different preconditioning approach and the addition of resolution length estimation. This approach is based on two key assumptions. First, the inversion needs to converge to the global minimum. Second, the model perturbation must be small enough to satisfy a linear approximation.

5.1 Posterior model covariance

In the linearized approximation described in Tarantola (2005), based on objective function (eq. 1), the posterior model covariance is determined by

$$\mathbf{C}_M = (\mathbf{J}^T \mathbf{C}_d^{-1} \mathbf{J} + \mathbf{C}_m^{-1})^{-1}. \quad (7)$$

Estimation of the prior, $\mathbf{C}_m$, is challenging, and we choose to be conservative, as detailed in Section 5.4.3. By assigning a relatively large prior, the posterior model covariance, $\mathbf{C}_M$, will be dominated by the term $\mathbf{J}^T \mathbf{C}_d^{-1} \mathbf{J}$, which reflects the data coverage of the inversion. Therefore, the uncertainty maps we provide in this section can be useful for future inversions that use GLAD-M35 as a starting model, because this prior can be built into the inversion.

Algorithm 3 of Liu *et al.* (2022) facilitates retrieval of the inverse Hessian entries from the preconditioned BFGS loops using a

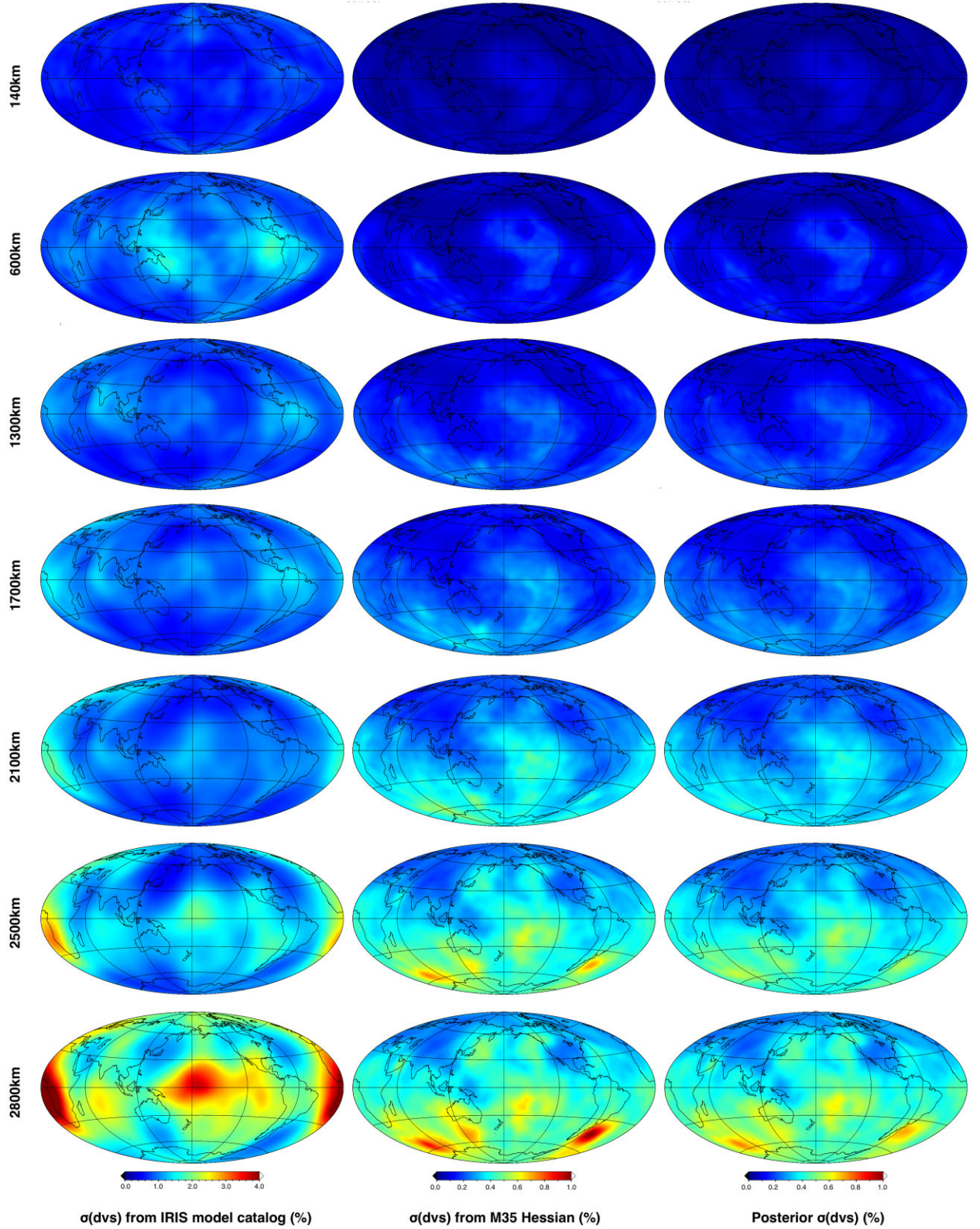


Figure 19. Map views of the shear wave speed perturbations prior, C_m (left-hand column), standard deviation from GLAD-M35 Hessian (middle column), and posterior, C_M , (right-hand column) at depths of 140 km, 600 km, 1300 km, 1700 km, 2100 km, 2500 km and 2800 km.

Randomized Singular Value Decomposition algorithm (Halko *et al.* 2011). The inverse Hessian is decomposed into the SVD form

$$\mathbf{H}^{-1} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T + \mathbf{B}_0, \quad (8)$$

where $\mathbf{\Lambda}$ and $\mathbf{V}$ denote the eigenvalues and eigenvectors, respectively. The dominant eigenvalues and eigenvectors are determined based on random probing. In expression (8), we define the

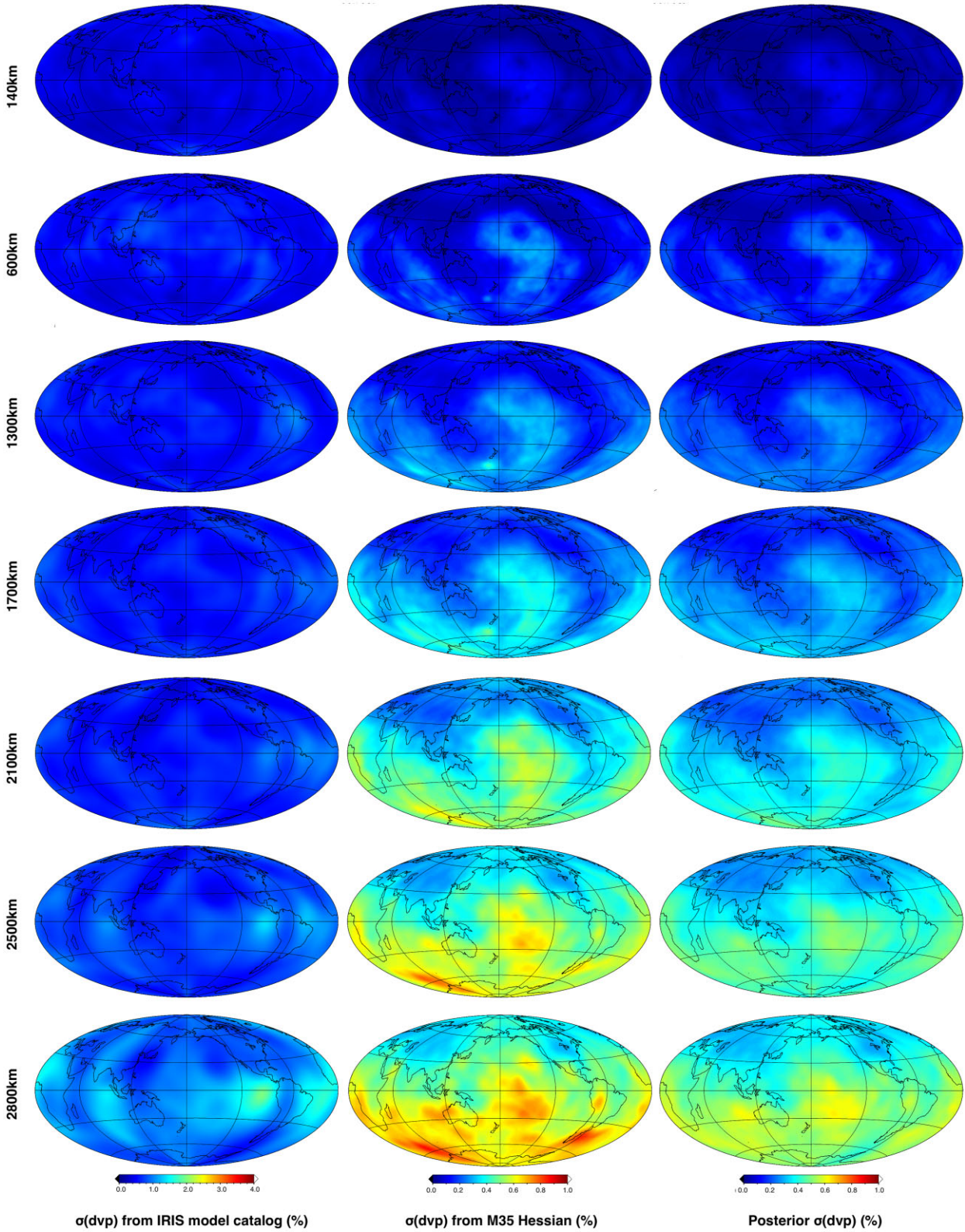


Figure 20. Same as Fig. 19, showing standard deviations of P wave speed perturbations.

regularization term $\mathbf{B}_0$ as

$$\mathbf{B}_0 \equiv (\lambda^{-1} \mathbf{H}_{\rho\rho} + \mathbf{C}_m^{-1})^{-1}. \quad (9)$$

Instead of defining $\mathbf{B}_0$ in terms of the diagonal Hessian contributions $\mathbf{H}_{\alpha\alpha}$, $\mathbf{H}_{\beta\beta}$ and $\mathbf{H}_{\rho\rho}$, as suggested by Liu *et al.* (2022), we

choose to use ray density, which is essentially $\mathbf{H}_{\rho\rho}$, for all parameters. The parameter λ scales $\mathbf{H}_{\rho\rho}$ and is determined by a line search. We observe that $\mathbf{B}_0$ captures regularization in the form of the inverse ray density as well as the prior $\mathbf{C}_m$. Thus, given $\mathbf{B}_0$ plus $\mathbf{V}$ and $\mathbf{\Lambda}$ from the BFGS and randomized SVD algorithms, the posterior $\mathbf{C}_M$

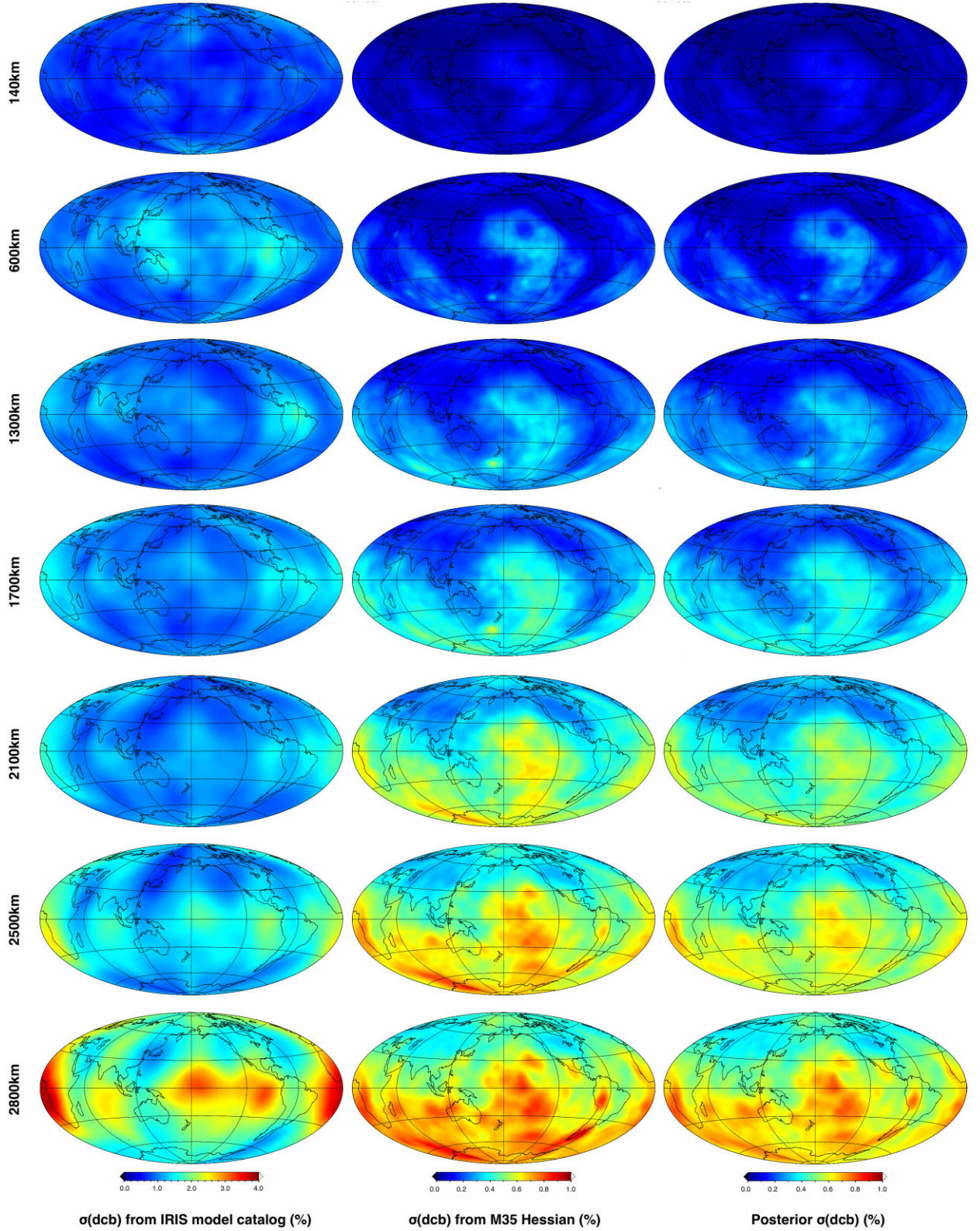


Figure 21. Same structure as Fig. 19, showing standard deviations of bulk sound speed perturbations.

may be calculated based on

$$\mathbf{C}_M = \mathbf{H}^{-1}, \quad (10)$$

which is the inverse Gauss–Newton Hessian determined based on randomized SVD.

Combining the BFGS and randomized SVD methods yields a low-rank representation of the inverse Hessian at an inexpensive cost. The approach is strictly based upon a standard FWI workflow, with the only additional requirement being the BFGS update history. The output inverse Hessian not only serves as an approximation of

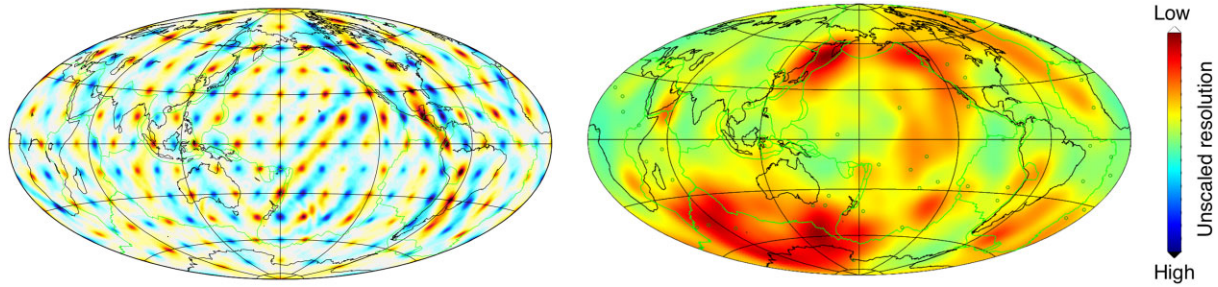


Figure 22. Point-spread function test for GLAD-M25 (left-hand panel) and unscaled resolution map in GLAD-M35 (right-hand panel) at 1200 km depth. Relatively low-resolution areas are consistent with ‘streaking’ in the point-spread-function map, specifically in western North America, South America and the Central Pacific. Low resolution in the colour bar means large resolution length.

posterior model covariance, but also gives us an idea of the scaling of each component, which enables us to properly scale $\mathbf{H}_{\alpha\alpha}$, $\mathbf{H}_{\beta\beta}$ and $\mathbf{H}_{\rho\rho}$ for future inversions.

5.2 Resolution length

The full picture of model uncertainty includes both the diagonal and off-diagonal terms of the Hessian. While a low-rank approximation of the Hessian or its inverse is good enough for estimating the diagonal terms, we do not have a sufficient number of iterations to support direct estimation of the off-diagonal terms. However, point-spread function tests show that the shape of the Hessian of point $\mathbf{x}$ at location $\mathbf{y}$ may be characterized as a Gaussian function (Fichtner & Trampert 2011):

$$\mathbf{H}(\mathbf{x}, \mathbf{y}) = \mathbf{M}(\mathbf{x}) \frac{\sqrt{\det \mathbf{C}(\mathbf{x})}}{(2\pi)^{3/2}} \exp \left[-\frac{1}{2}(\mathbf{y} - \mathbf{x})^T \mathbf{C}(\mathbf{x}) (\mathbf{y} - \mathbf{x}) \right]. \quad (11)$$

The symmetric matrix $\mathbf{C}(\mathbf{x})$ describes the width and orientation of the Hessian centred at $\mathbf{x}$. The weight $\mathbf{M}(\mathbf{x}) \approx \int \mathbf{H}(\mathbf{x}, \mathbf{y}) d^3\mathbf{y}$ is equal to the integral over the Hessian for a point perturbation at location $\mathbf{x}$. The resolution length in a specific direction $\mathbf{e}$ may be defined as

$$l_e \equiv \frac{1}{\sqrt{\mathbf{e}^T \mathbf{C} \mathbf{e}}}. \quad (12)$$

By examining the diagonal terms of expression (11), setting $\mathbf{x} = \mathbf{y}$, we notice that l_e is directly determined by the ratio between the diagonal Hessian values $\mathbf{H}(\mathbf{x}, \mathbf{x})$ and the spatial integral $\mathbf{M}(\mathbf{x})$. We have approximated $\mathbf{H}^{-1}(\mathbf{x}, \mathbf{x})$ with the BFGS update history for uncertainty, and the same algorithm can also be used to approximate $\mathbf{H}(\mathbf{x}, \mathbf{x})$ with just a small change in formula (De Sterck & Howse 2018). Obtaining $\mathbf{M}(\mathbf{x})$ is even more straightforward because we can directly probe the BFGS update history with $[1, 1, \dots, 1]$. As we do not expect a good estimate of individual off-diagonal terms, the scale length l_e obtained in this way also requires an extra scaling step. We scale l_e based on the results of a point-spread function test.

5.3 Summary

Through the BFGS update history, we can approximate the product of the Hessian or the inverse Hessian with any vector. By probing with random vectors, we obtain the essential features of Hessian or its inverse based on randomized singular value decomposition. We approximate the inverse Hessian’s diagonal terms for the uncertainty maps. For the resolution length, we approximate the diagonal Hessian and the integral of the Hessian.

5.4 Application to GLAD-M35

In this section, we apply uncertainty quantification and resolution length estimation to model GLAD-M35. We discuss the number of required eigenvalues, data uncertainty, and a practical way to estimate the prior.

5.4.1 Number of required eigenvalues

The Hessian and its inverse are probed by random vectors to construct the relative and smoothed shape of three vectors: the spatial integral of the Hessian, the diagonal terms of the Hessian, and the diagonal terms of the inverse Hessian. This requires much less information than constructing the full Hessian. To illustrate this, we approximate the diagonal inverse Hessian by probing the BFGS history with 3, 5, 8 and 10 random vectors (Fig. 17). We see that the main features of the diagonal Hessian gradually stabilize as the number of vectors increases. The difference between using 8 and 10 vectors is small. Therefore, we use ten probing vectors.

5.4.2 Data uncertainty

Fig. 18 shows the distribution of multi-taper phase misfit values (eq. 2). The standard deviation of the phase misfit is 1.09, which is very close to a standard normal distribution. This shows that the estimated data uncertainty in the multi-taper misfit is reasonable.

5.4.3 Prior information and posterior uncertainty

We estimate the prior model uncertainty, $\mathbf{C}_M$, by analysing the statistics of 13 global models, namely, GyPSuM (Simmons *et al.* 2010), HMSL (Houser *et al.* 2008), GLAD-M25 (Lei *et al.* 2020), REVEAL (Thrastarson *et al.* 2024), SALSA3D (Ballard *et al.* 2016), SAVANI-US (Porritt *et al.* 2021), SEISGLOB2 (Durand *et al.* 2017), SEMUCB-WM1 (French & Romanowicz 2014), SGLOBE-rani (Chang *et al.* 2014), SP12RTS (Koelemeijer *et al.* 2016), SPani (Tesoniero *et al.* 2015), SPiRaL-1.4 (Simmons *et al.* 2021) and TX2019 (Lu *et al.* 2019). While it is easy to calculate standard deviations given these models, their resolution lengths are much harder to estimate, and the standard deviations are not meaningful without this. Furthermore, given the uncertainty in these models, the linear approximation can hardly hold. Nevertheless, we estimate a prior by assuming that the resolution length of these models increases linearly from 50 to 750 km from the top to the bottom of the mantle. The resulting prior, $\mathbf{C}_M$, is shown in the first column of Figs 19, 20 and 21, the contributions from GLAD-M35 is shown in the second column, and the posterior, $\mathbf{C}_M$, is shown in the third

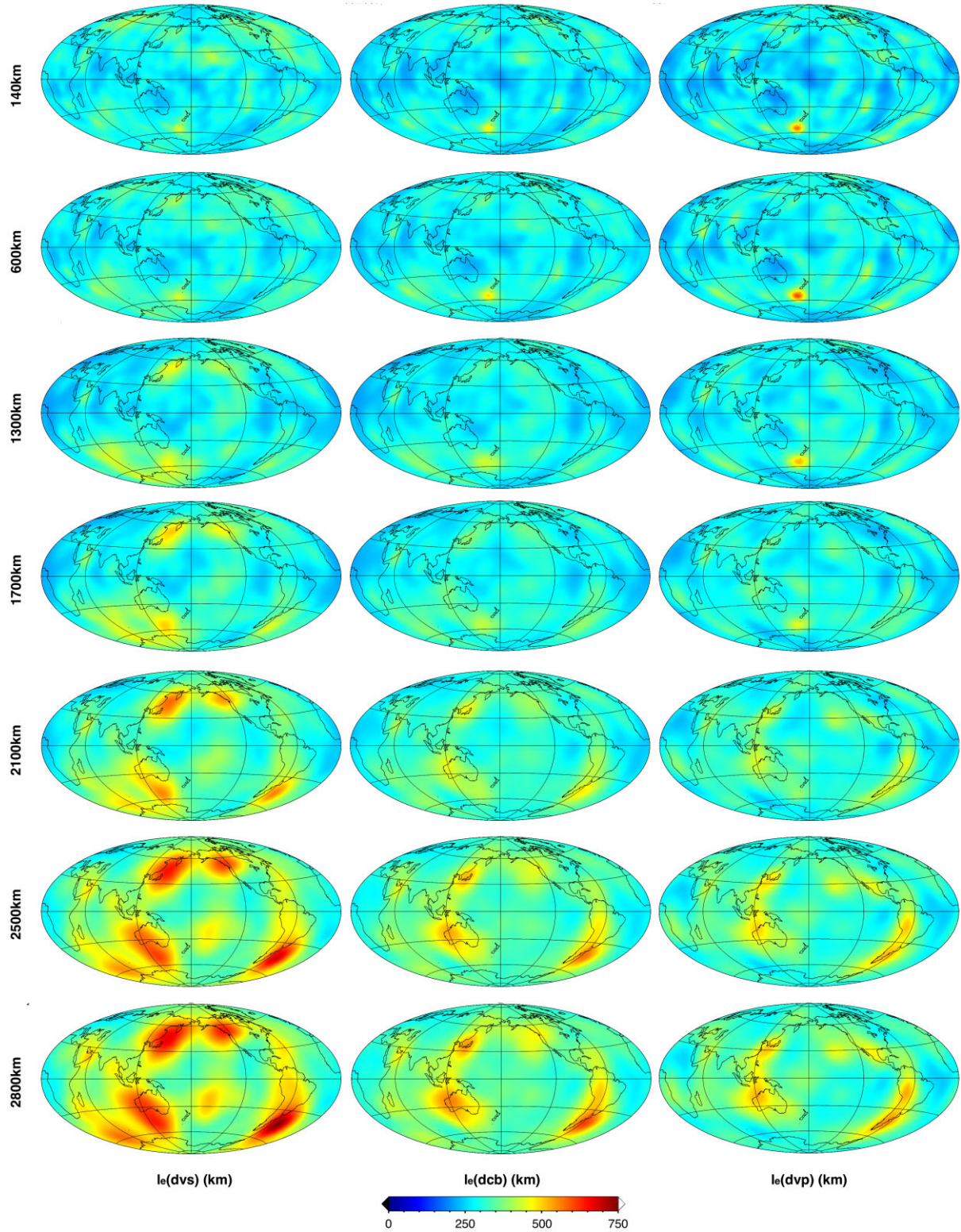


Figure 23. Horizontal resolution map for model GLAD-M35 at depths of 140 km, 600 km, 1300 km, 1700 km, 2100 km, 2500 km and 2800 km, showing resolution length (eq. 12) (left-hand column), bulk sound speed resolution length (middle column), and compressional wave speed resolution length (right-hand column). The resolution length is scaled based on the PSF test for GLAD-M25 shown in Fig. 22 (left-hand panel).

column. In general, the prior uncertainty increases with depth and the S wave speed has a larger uncertainty than the P wave speed because its overall perturbations are stronger. We also notice that the bulk sound speed has a relatively large uncertainty compared to

the result from GLAD-M35 (middle columns). This may be because many models were obtained without joint P and S constraints.

Prior information, C_m , does not play a significant role in the posterior, C_M , which is shown in the third column of Figs 19, 20

and 21. The work of van Herwaarden *et al.* (2020), obtaining a 3-D model from a 1-D initial model, also suggests that prior information may not be essential for global adjoint tomography. This does not mean that prior information is not relevant because carefully selecting a prior has the potential to reduce posterior uncertainty. However, our results suggest that a few thousand earthquakes can resolve a global model at the resolution of GLAD-M35, even with a conservative prior.

5.4.4 Resolution length

By comparing a point-spread function for GLAD-M25 with an (unscaled) resolution length estimate computed based on eq. (12) in Fig. 22, we observe that the low-resolution areas are somewhat consistent, including relatively low resolution (larger resolution length values) in the western United States and South America, which is expected because resolution is inversely proportional to uncertainty. The relatively low resolution in the area south of Australia is not revealed in GLAD-M25's PSF test, which is likely due to the new data in the GLAD-M35 data set. Based on the figure, we scale the resolution map such that the mean horizontal resolution length at 1200 km is 250 km. The final resolution map is shown in Fig. 23. This result does not account for the non-linearity of the iterative inversion but serves as a conservative estimate of the range of impact that a standard deviation point perturbation has on the model. These resolution lengths justify the range of values used in the determination of the prior in Section 5.4.3.

Unlike the uncertainty map, the resolution length map involves some approximations and subjective scaling, which is more suitable for certain quantitative analyses, such as calculating R values. The resolution map, however, should only be used qualitatively. In a scenario where the model has the same resolution everywhere, the resolution length should be inversely proportional to the standard deviation. Hence, regions in the map with low uncertainty and short resolution lengths are more reliable, whereas regions with high uncertainty and long resolution lengths are not well-resolved. By examining the uncertainty and resolution maps together, we can observe that the continents in the Northern Hemisphere are generally better resolved. The two most questionable regions are the Southern Ocean and the southern part of South America. This suggests the need to collect and process data to better constrain these regions in the future.

6 DISCUSSION AND CONCLUSION

We used the second-generation global adjoint tomography model GLAD-M25 as a starting model and performed ten iterations using improved data processing and measurement techniques to create model GLAD-M35 with uncertainty quantification. GLAD-M35 provides more detailed images of the deep mantle and highlights the importance of window selection criteria and a new P -wave category. A significant change in bulk sound speed in the deep mantle significantly reduces the data misfit of the related P -wave phases and aligns well with geodynamic constraints. The model has converged for three reasons: there was only a minor misfit reduction after the third iteration, the centre of the traveltimes anomaly histograms in all wave categories approaches zero, and there were no significant model updates except in regions with new data.

Quantifying uncertainty is a challenging task that involves making multiple assumptions. After studying the data statistics and model update values, we have concluded that the inversion has

converged, which satisfies one of the requirements for linearized uncertainty quantification. However, we cannot guarantee that there are no regions that have fallen into a local minimum, especially in the deep mantle, due to uneven data coverage. The scaling of the inverse Hessian requires caution, as the non-linearity of the forward problem can lead to inaccuracies in the line search. Nevertheless, we were able to provide sensible overall uncertainty and resolution length estimates of the model through a mathematical approach that does not rely on experience or much prior information. The results indicate directions for further improvements in data processing, objective functions, and pre-conditioners, and can serve as a starting point for future Bayesian-based inversions.

The GLAD series of models has been updated to its final version, GLAD-M35. To make significant further updates, it is necessary to include more earthquakes or higher frequencies while accommodating azimuthal anisotropy and attenuation. Unfortunately, increasing the database would require a large amount of storage space, making it an unappealing prospect. Therefore, we plan to use new techniques to reduce the computational cost of global tomography after GLAD-M35. These techniques include Model-Order Reduction (e.g. Hawkins *et al.* 2023) and Source Encoding (e.g. Cui *et al.* 2023).

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DATA AVAILABILITY

The GLAD-M35 model is available from the EarthScope Earth Model Collaboration webpage (Cui *et al.* 2024) at a resolution for visualization purposes. The full model for numerical simulations and other necessary data will be provided upon request by the corresponding author.

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APPENDIX A: MODEL GLAD-M25

For reference and comparison, map views of our second-generation global adjoint tomography model GLAD-M25 are shown in Fig. A1, complementing Fig. 9.

APPENDIX B: 1-D MEAN PROFILES OF GLAD-M35

The 1-D mean V_p , V_s and V_p/V_s profiles of GLAD-M35, GLAD-M25 and 1-D reference model STW105 are shown in Fig. A2.

APPENDIX C: R PROFILE CALCULATION

Since directly calculating the S -to- P wave speed perturbation ratio (eq. 6) may result in division by zero, we calculate it in a least-squares sense. As described in Ishii & Tromp (2001), we define an objective function

$$f(R) = \sum_{i=1}^N \frac{[(\delta \log V_s)_i - R(\delta \log V_p)_i]^2}{(\sigma_{V_s})_i^2 + R^2(\sigma_{V_p})_i^2}, \quad (\text{C1})$$

which, at a given depth, involves a sum over N wave speed perturbations $\delta \log V_s$ and $\delta \log V_p$ with associated standard deviations σ_{V_s} and σ_{V_p} . We seek to determine the value of R that minimizes f . After exchanging V_p and V_s in eq. (C1), we repeat the calculation, which yields an estimate for $1/R$, and average the two R values. For models without uncertainty quantification, including GLAD-M25, LOWE and REVEAL, we define $f(R)$ as

$$f(R) = \sum_i [(\delta \log V_s)_i - R(\delta \log V_p)_i]^2, \quad (\text{C2})$$

and determine the value of R which minimizes f .

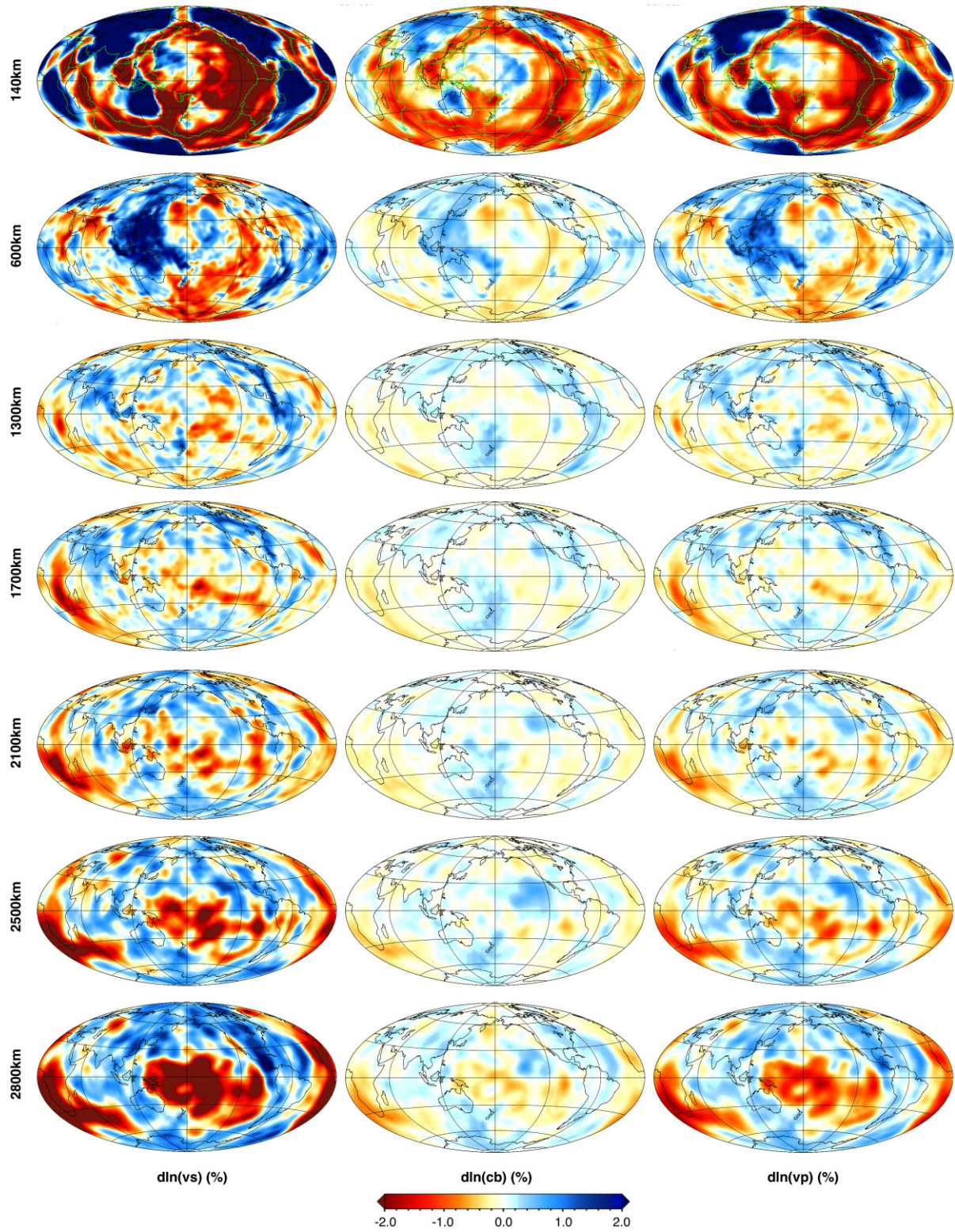


Figure A1. Map views of second-generation global adjoint tomography model GLAD-M25 at depths of 140 km, 600 km, 1300 km, 1700 km, 2100 km, 2500 km and 2800 km, showing shear wave speed perturbations (left-hand column), bulk sound speed perturbations (middle column) and compressional wave speed perturbations (right-hand column).

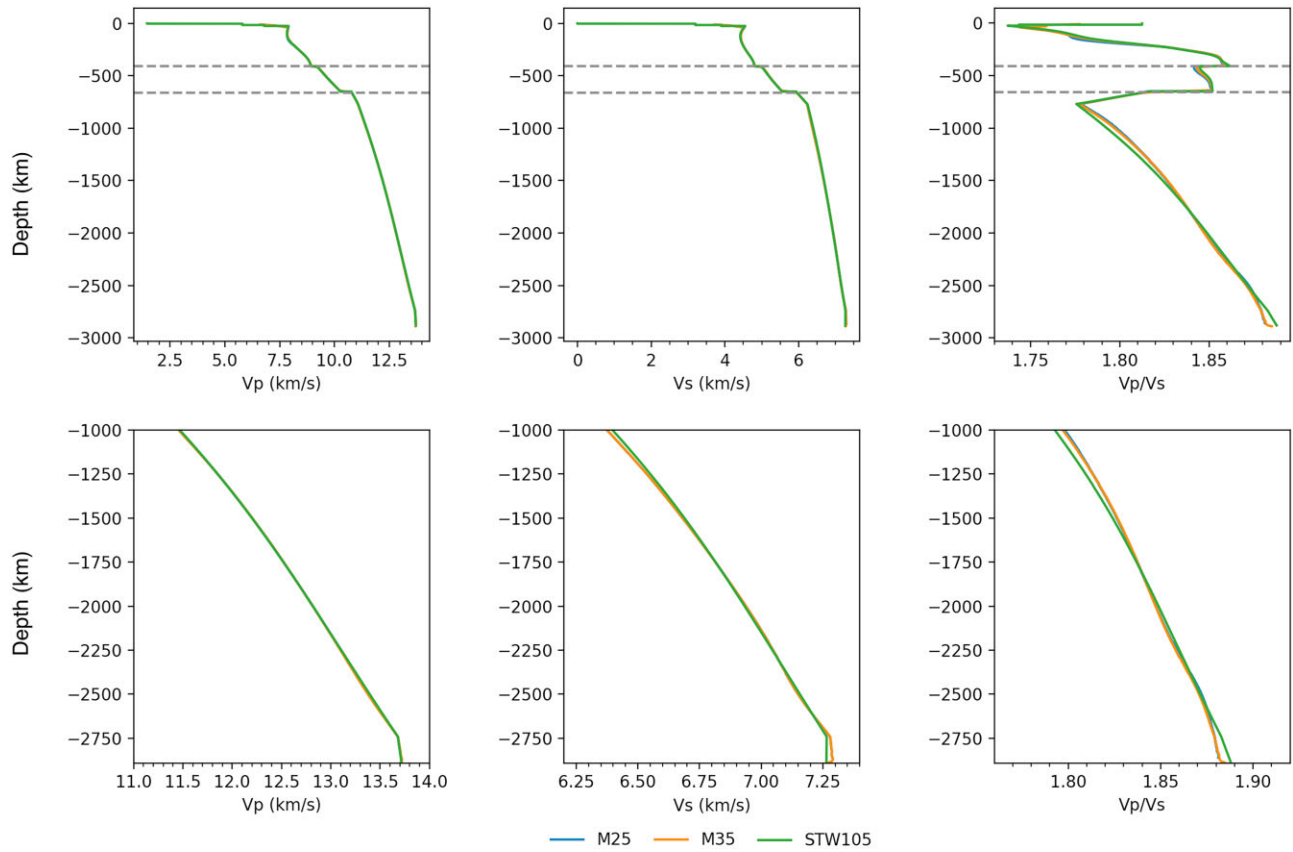


Figure A2. 1-D mean profiles of V_p , V_s and V_p/V_s (top row) and their zoomed-in lower-mantle sections (bottom row) for models GLAD-M25 (blue), GLAD-M35 (orange) and STW105 (green).